

mp4

April 18, 2025

```
[1]: # # you will be prompted with a window asking to grant permissions
# from google.colab import drive
# drive.mount("/content/drive")

[2]: # # fill in the path in your Google Drive in the string below. Note: do not
# ↪ escape slashes or spaces
# import os
# datadir = "/content/assignment4"
# if not os.path.exists(datadir):
#     !ln -s "/content/drive/MyDrive/MP4/" $datadir # TODO: Fill your Assignment
# ↪ 4 path
# os.chdir(datadir)
# !pwd

[3]: # NOTE: Loading data from Google Drive is VERY SLOW. Therefore, we download the
# ↪ data to a local storage
# specified by VOC_PATH.
# This means the data would have be re-downloaded everytime.
# Change this path if you are running this script locally
VOC_PATH="data"

# !chmod u+x ./download_data.sh
# !sed -i 's/\r//g' ./download_data.sh
# !cat ./download_data.sh
# !bash ./download_data.sh $VOC_PATH

[4]: import os
import random

import cv2
import numpy as np

import torch
from torch.utils.data import DataLoader

from src.resnet_yolo import resnet50
import yolo_loss
```

```

from src.dataset import VocDetectorDataset
from src.eval_voc import evaluate
from src.predict import predict_image
from src.config import VOC_CLASSES, COLORS
from kaggle_submission import output_submission_csv

import matplotlib.pyplot as plt
import collections

import time

%matplotlib inline
%load_ext autoreload
%autoreload 2

```

0.1 Initialization

```

[5]: device = torch.device("cuda:0" if torch.cuda.is_available() else "mps" if torch.
    ↪mps.is_available() else "cpu")
device

```

```

[5]: device(type='mps')

```

```

[6]: # YOLO network hyperparameters
B = 2 # number of bounding box predictions per cell
S = 14 # width/height of network output grid (larger than 7x7 from paper since ↪
    ↪we use a different network)

```

0.2 Start from here if you modified yolo_loss.py and wish to retrain

```

[7]: # We recommend tuning only the batch size; the remaining hyperparameters should ↪
    ↪work well
# with the default values. However, you are free to experiment with them.

learning_rate = 0.001
num_epochs = 50
batch_size = 12

# Yolo loss component coefficients (as given in Yolo v1 paper)
lambda_coord = 5
lambda_noobj = 0.5

```

0.3 Reading Pascal Data

Since Pascal is a small dataset (5000 in train+val) we have combined the train and val splits to train our detector. This is not typically a good practice, but we will make an exception in this case to be able to get reasonable detection results with a comparatively small object detection dataset.

The train dataset loader also using a variety of data augmentation techniques including random shift, scaling, crop, and flips. Data augmentation is slightly more complicated for detection datasets since the bounding box annotations must be kept consistent throughout the transformations.

Since the output of the detector network we train is an $S \times S \times (B \times 5 + C)$, we use an encoder to convert the original bounding box coordinates into relative grid bounding box coordinates corresponding to the expected output. We also use a decoder which allows us to convert the opposite direction into image coordinate bounding boxes.

```
[8]: file_root_train = os.path.join(VOC_PATH, 'VOCdevkit_2007/VOC2007/JPEGImages/')
      annotation_file_train = 'data/voc2007.txt'

      train_dataset = □
        ↪VocDetectorDataset(root_img_dir=file_root_train,dataset_file=annotation_file_train,train=True,
        ↪S=S)
      train_loader = □
        ↪DataLoader(train_dataset,batch_size=batch_size,shuffle=True,num_workers=2)
      print('Loaded %d train images' % len(train_dataset))
```

Initializing dataset

Loaded 5011 train images

```
[9]: file_root_test = os.path.join(VOC_PATH, 'VOCdevkit_2007/VOC2007test/JPEGImages/'
      ↪')
      annotation_file_test = 'data/voc2007test.txt'

      test_dataset = □
        ↪VocDetectorDataset(root_img_dir=file_root_test,dataset_file=annotation_file_test,train=False,
        ↪S=S)
      test_loader = □
        ↪DataLoader(test_dataset,batch_size=batch_size,shuffle=False,num_workers=2)
      print('Loaded %d test images' % len(test_dataset))
```

Initializing dataset

Loaded 4950 test images

```
[10]: data = train_dataset[0]
```

0.4 Initializing the network

To implement Yolo we will rely on a pretrained classifier as the backbone for our detection network. PyTorch offers a variety of models which are pretrained on ImageNet in the `torchvision.models` package. In particular, we will use the ResNet50 architecture as a base for our detector. This is different from the base architecture in the Yolo paper and also results in a different output grid size (14x14 instead of 7x7).

Models are typically pretrained on ImageNet since the dataset is very large (> 1 million images) and widely used. The pretrained model provides a very useful weight initialization for our detector, so that the network is able to learn quickly and effectively.

```
[11]: load_network_path = None
pretrained = True

# use to load a previously trained network
if load_network_path is not None:
    print('Loading saved network from {}'.format(load_network_path))
    net = resnet50().to(device)
    net.load_state_dict(torch.load(load_network_path))
else:
    print('Load pre-trained model')
    net = resnet50(pretrained=pretrained).to(device)
```

Load pre-trained model

0.5 Set up training tools

```
[12]: from importlib import reload
reload(yolo_loss) # update the import if the implementation has changed

criterion = yolo_loss.YoloLoss(S, B, lambda_coord, lambda_noobj)
optimizer = torch.optim.SGD(net.parameters(), lr=learning_rate, momentum=0.9,
↪weight_decay=5e-4)
```

0.6 Train detector

```
[25]: best_test_loss = np.inf
learning_rate = learning_rate

os.makedirs('./checkpoints', exist_ok=True)
for epoch in range(num_epochs):
    net.train()

    # Update learning rate late in training
    if epoch == 30 or epoch == 40:
        learning_rate /= 10.0

    for param_group in optimizer.param_groups:
        param_group['lr'] = learning_rate

    print('\n\nStarting epoch %d / %d' % (epoch + 1, num_epochs))
    print('Learning Rate for this epoch: {}'.format(learning_rate))

    total_loss = collections.defaultdict(int)

    start_iteration_time = time.time()
    for i, data in enumerate(train_loader):
        data = (item.to(device) for item in data)
        images, target_boxes, target_cls, has_object_map = data
```

```

    pred = net(images)
    loss_dict = criterion(pred, target_boxes, target_cls, has_object_map)
    for key in loss_dict:
        total_loss[key] += loss_dict[key].item()

    optimizer.zero_grad()
    loss_dict['total_loss'].backward()
    optimizer.step()

    if (i+1) % 50 == 0:
        outstring = 'Epoch [%d/%d], Iter [%d/%d], Loss: ' % ((epoch+1,
↪num_epochs, i+1, len(train_loader)))
        outstring += ', '.join( "%s=%.3f" % (key[:-5], val / (i+1)) for
↪key, val in total_loss.items() )
        print(outstring)
    print('Time for epoch: %.2f seconds' % (time.time() - start_iteration_time))

    # evaluate the network on the test data
    if (epoch + 1) % 5 == 0:
        test_aps = evaluate(net, test_dataset_file=annotation_file_test,
↪img_root=file_root_test)
        print(epoch, test_aps)
        with torch.no_grad():
            test_loss = 0.0
            net.eval()
            for i, data in enumerate(test_loader):
                data = (item.to(device) for item in data)
                images, target_boxes, target_cls, has_object_map = data

                pred = net(images)
                loss_dict = criterion(pred, target_boxes, target_cls,
↪has_object_map)
                test_loss += loss_dict['total_loss'].item()
            test_loss /= len(test_loader)

            if best_test_loss > test_loss:
                best_test_loss = test_loss
                print('Updating best test loss: %.5f' % best_test_loss)
                torch.save(net.state_dict(), 'checkpoints/best_detector.pth')

            if (epoch+1) in [5, 10, 20, 30, 40]:
                torch.save(net.state_dict(), 'checkpoints/detector_epoch_%d.pth' %
↪(epoch+1))

        torch.save(net.state_dict(), 'checkpoints/detector.pth')

```

Starting epoch 1 / 50

Learning Rate for this epoch: 0.001

Epoch [1/50], Iter [50/418], Loss: total=24.574, reg=48.614,
containing_obj=3.432, no_obj=149.910, cls=92.929
Epoch [1/50], Iter [100/418], Loss: total=16.512, reg=42.429,
containing_obj=4.417, no_obj=77.594, cls=73.710
Epoch [1/50], Iter [150/418], Loss: total=13.513, reg=39.547,
containing_obj=5.047, no_obj=52.951, cls=64.607
Epoch [1/50], Iter [200/418], Loss: total=11.821, reg=37.052,
containing_obj=5.514, no_obj=40.436, cls=58.849
Epoch [1/50], Iter [250/418], Loss: total=10.733, reg=35.531,
containing_obj=5.879, no_obj=32.831, cls=54.556
Epoch [1/50], Iter [300/418], Loss: total=9.864, reg=33.971,
containing_obj=6.088, no_obj=27.707, cls=50.604
Epoch [1/50], Iter [350/418], Loss: total=9.229, reg=32.894,
containing_obj=6.332, no_obj=24.012, cls=47.506
Epoch [1/50], Iter [400/418], Loss: total=8.783, reg=32.345,
containing_obj=6.558, no_obj=21.217, cls=45.276
Time for epoch: 197.86 seconds
Updating best test loss: 5.32967

Starting epoch 2 / 50

Learning Rate for this epoch: 0.001

Epoch [2/50], Iter [50/418], Loss: total=5.199, reg=25.905,
containing_obj=8.081, no_obj=1.474, cls=26.928
Epoch [2/50], Iter [100/418], Loss: total=5.156, reg=26.312,
containing_obj=8.313, no_obj=1.421, cls=25.828
Epoch [2/50], Iter [150/418], Loss: total=5.071, reg=25.826,
containing_obj=8.354, no_obj=1.375, cls=25.300
Epoch [2/50], Iter [200/418], Loss: total=5.055, reg=25.644,
containing_obj=8.388, no_obj=1.334, cls=25.296
Epoch [2/50], Iter [250/418], Loss: total=5.030, reg=25.632,
containing_obj=8.458, no_obj=1.297, cls=24.976
Epoch [2/50], Iter [300/418], Loss: total=5.029, reg=25.769,
containing_obj=8.548, no_obj=1.264, cls=24.763
Epoch [2/50], Iter [350/418], Loss: total=4.991, reg=25.657,
containing_obj=8.607, no_obj=1.235, cls=24.394
Epoch [2/50], Iter [400/418], Loss: total=4.913, reg=25.297,
containing_obj=8.568, no_obj=1.209, cls=23.878
Time for epoch: 194.51 seconds
Updating best test loss: 4.71994

Starting epoch 3 / 50

Learning Rate for this epoch: 0.001

Epoch [3/50], Iter [50/418], Loss: total=4.535, reg=23.973,

containing_obj=9.087, no_obj=0.983, cls=20.373
Epoch [3/50], Iter [100/418], Loss: total=4.512, reg=24.032,
containing_obj=9.066, no_obj=0.973, cls=20.069
Epoch [3/50], Iter [150/418], Loss: total=4.512, reg=23.994,
containing_obj=9.089, no_obj=0.965, cls=20.092
Epoch [3/50], Iter [200/418], Loss: total=4.464, reg=23.743,
containing_obj=9.030, no_obj=0.960, cls=19.839
Epoch [3/50], Iter [250/418], Loss: total=4.485, reg=24.006,
containing_obj=9.020, no_obj=0.955, cls=19.833
Epoch [3/50], Iter [300/418], Loss: total=4.488, reg=24.062,
containing_obj=9.045, no_obj=0.951, cls=19.797
Epoch [3/50], Iter [350/418], Loss: total=4.466, reg=23.980,
containing_obj=9.020, no_obj=0.948, cls=19.648
Epoch [3/50], Iter [400/418], Loss: total=4.465, reg=23.970,
containing_obj=9.040, no_obj=0.946, cls=19.630
Time for epoch: 194.89 seconds
Updating best test loss: 4.40350

Starting epoch 4 / 50
Learning Rate for this epoch: 0.001
Epoch [4/50], Iter [50/418], Loss: total=4.093, reg=22.149,
containing_obj=8.989, no_obj=0.955, cls=17.019
Epoch [4/50], Iter [100/418], Loss: total=4.251, reg=22.960,
containing_obj=9.028, no_obj=0.961, cls=18.062
Epoch [4/50], Iter [150/418], Loss: total=4.235, reg=23.121,
containing_obj=9.082, no_obj=0.969, cls=17.645
Epoch [4/50], Iter [200/418], Loss: total=4.290, reg=23.545,
containing_obj=9.156, no_obj=0.983, cls=17.798
Epoch [4/50], Iter [250/418], Loss: total=4.273, reg=23.445,
containing_obj=9.147, no_obj=0.995, cls=17.684
Epoch [4/50], Iter [300/418], Loss: total=4.257, reg=23.384,
containing_obj=9.132, no_obj=1.008, cls=17.556
Epoch [4/50], Iter [350/418], Loss: total=4.236, reg=23.166,
containing_obj=9.086, no_obj=1.010, cls=17.574
Epoch [4/50], Iter [400/418], Loss: total=4.223, reg=23.090,
containing_obj=9.119, no_obj=1.012, cls=17.453
Time for epoch: 194.12 seconds
Updating best test loss: 4.14806

Starting epoch 5 / 50
Learning Rate for this epoch: 0.001
Epoch [5/50], Iter [50/418], Loss: total=3.830, reg=21.039,
containing_obj=8.791, no_obj=1.095, cls=15.029
Epoch [5/50], Iter [100/418], Loss: total=3.964, reg=21.817,
containing_obj=9.185, no_obj=1.073, cls=15.495
Epoch [5/50], Iter [150/418], Loss: total=4.009, reg=21.944,

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containing_obj=9.293, no_obj=1.080, cls=15.793
Epoch [5/50], Iter [200/418], Loss: total=4.023, reg=22.050,
containing_obj=9.316, no_obj=1.090, cls=15.818
Epoch [5/50], Iter [250/418], Loss: total=3.990, reg=21.834,
containing_obj=9.257, no_obj=1.095, cls=15.690
Epoch [5/50], Iter [300/418], Loss: total=3.973, reg=21.809,
containing_obj=9.221, no_obj=1.098, cls=15.553
Epoch [5/50], Iter [350/418], Loss: total=3.962, reg=21.728,
containing_obj=9.209, no_obj=1.101, cls=15.507
Epoch [5/50], Iter [400/418], Loss: total=3.951, reg=21.757,
containing_obj=9.201, no_obj=1.108, cls=15.347
Time for epoch: 194.58 seconds
---Evaluate model on test samples---
100%|      | 4950/4950 [01:23<00:00, 59.24it/s]
---class aeroplane ap 0.03373345644813789---
---class bicycle ap 0.19277853665142153---
---class bird ap 0.0--- (no predictions for this class)
---class boat ap 0.0--- (no predictions for this class)
---class bottle ap 0.0--- (no predictions for this class)
---class bus ap 0.0--- (no predictions for this class)
---class car ap 0.28169322909411665---
---class cat ap 0.0---
---class chair ap 0.04246804964140703---
---class cow ap 0.0--- (no predictions for this class)
---class diningtable ap 0.0--- (no predictions for this class)
---class dog ap 0.016700749829584187---
---class horse ap 0.026907001044932082---
---class motorbike ap 0.0--- (no predictions for this class)
---class person ap 0.13291835702278842---
---class pottedplant ap 0.009445875305250305---
---class sheep ap 0.0--- (no predictions for this class)
---class sofa ap 0.0--- (no predictions for this class)
---class train ap 0.0--- (no predictions for this class)
---class tvmonitor ap 0.013798701298701298---
---map 0.03752219781681697---
4 [np.float64(0.03373345644813789), np.float64(0.19277853665142153), 0.0, 0.0,
0.0, 0.0, np.float64(0.28169322909411665), np.float64(0.0),
np.float64(0.04246804964140703), 0.0, 0.0, np.float64(0.016700749829584187),
np.float64(0.026907001044932082), 0.0, np.float64(0.13291835702278842),
np.float64(0.009445875305250305), 0.0, 0.0, 0.0,
np.float64(0.013798701298701298)]
Updating best test loss: 3.94850

Starting epoch 6 / 50
Learning Rate for this epoch: 0.001
Epoch [6/50], Iter [50/418], Loss: total=3.727, reg=21.288,

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containing_obj=9.233, no_obj=1.172, cls=13.033
Epoch [6/50], Iter [100/418], Loss: total=3.802, reg=21.191,
containing_obj=9.499, no_obj=1.156, cls=13.777
Epoch [6/50], Iter [150/418], Loss: total=3.838, reg=21.383,
containing_obj=9.366, no_obj=1.155, cls=14.154
Epoch [6/50], Iter [200/418], Loss: total=3.780, reg=21.057,
containing_obj=9.298, no_obj=1.150, cls=13.855
Epoch [6/50], Iter [250/418], Loss: total=3.753, reg=20.858,
containing_obj=9.225, no_obj=1.153, cls=13.803
Epoch [6/50], Iter [300/418], Loss: total=3.734, reg=20.798,
containing_obj=9.216, no_obj=1.154, cls=13.636
Epoch [6/50], Iter [350/418], Loss: total=3.739, reg=20.812,
containing_obj=9.228, no_obj=1.152, cls=13.674
Epoch [6/50], Iter [400/418], Loss: total=3.751, reg=20.933,
containing_obj=9.252, no_obj=1.149, cls=13.675
Time for epoch: 194.48 seconds
Updating best test loss: 3.74622

Starting epoch 7 / 50
Learning Rate for this epoch: 0.001
Epoch [7/50], Iter [50/418], Loss: total=3.556, reg=20.334,
containing_obj=9.315, no_obj=1.125, cls=11.904
Epoch [7/50], Iter [100/418], Loss: total=3.643, reg=20.789,
containing_obj=9.395, no_obj=1.140, cls=12.388
Epoch [7/50], Iter [150/418], Loss: total=3.586, reg=20.514,
containing_obj=9.113, no_obj=1.164, cls=12.245
Epoch [7/50], Iter [200/418], Loss: total=3.615, reg=20.651,
containing_obj=9.241, no_obj=1.169, cls=12.320
Epoch [7/50], Iter [250/418], Loss: total=3.610, reg=20.588,
containing_obj=9.220, no_obj=1.182, cls=12.336
Epoch [7/50], Iter [300/418], Loss: total=3.596, reg=20.531,
containing_obj=9.200, no_obj=1.184, cls=12.233
Epoch [7/50], Iter [350/418], Loss: total=3.574, reg=20.433,
containing_obj=9.199, no_obj=1.189, cls=12.071
Epoch [7/50], Iter [400/418], Loss: total=3.565, reg=20.363,
containing_obj=9.210, no_obj=1.197, cls=12.016
Time for epoch: 193.63 seconds
Updating best test loss: 3.55244

Starting epoch 8 / 50
Learning Rate for this epoch: 0.001
Epoch [8/50], Iter [50/418], Loss: total=3.531, reg=20.621,
containing_obj=9.424, no_obj=1.256, cls=11.069
Epoch [8/50], Iter [100/418], Loss: total=3.456, reg=20.271,
containing_obj=9.273, no_obj=1.237, cls=10.695
Epoch [8/50], Iter [150/418], Loss: total=3.376, reg=19.683,

containing_obj=9.084, no_obj=1.246, cls=10.503
Epoch [8/50], Iter [200/418], Loss: total=3.396, reg=19.743,
containing_obj=9.166, no_obj=1.265, cls=10.574
Epoch [8/50], Iter [250/418], Loss: total=3.380, reg=19.684,
containing_obj=9.136, no_obj=1.267, cls=10.472
Epoch [8/50], Iter [300/418], Loss: total=3.390, reg=19.828,
containing_obj=9.180, no_obj=1.282, cls=10.394
Epoch [8/50], Iter [350/418], Loss: total=3.381, reg=19.728,
containing_obj=9.176, no_obj=1.288, cls=10.383
Epoch [8/50], Iter [400/418], Loss: total=3.366, reg=19.651,
containing_obj=9.157, no_obj=1.299, cls=10.285
Time for epoch: 193.29 seconds
Updating best test loss: 3.42392

Starting epoch 9 / 50

Learning Rate for this epoch: 0.001

Epoch [9/50], Iter [50/418], Loss: total=3.116, reg=18.350,
containing_obj=8.662, no_obj=1.365, cls=9.011
Epoch [9/50], Iter [100/418], Loss: total=3.176, reg=18.758,
containing_obj=9.071, no_obj=1.387, cls=8.893
Epoch [9/50], Iter [150/418], Loss: total=3.169, reg=18.599,
containing_obj=8.988, no_obj=1.391, cls=9.055
Epoch [9/50], Iter [200/418], Loss: total=3.211, reg=18.885,
containing_obj=9.049, no_obj=1.395, cls=9.205
Epoch [9/50], Iter [250/418], Loss: total=3.197, reg=18.882,
containing_obj=8.938, no_obj=1.404, cls=9.145
Epoch [9/50], Iter [300/418], Loss: total=3.215, reg=19.130,
containing_obj=9.009, no_obj=1.411, cls=9.029
Epoch [9/50], Iter [350/418], Loss: total=3.207, reg=19.051,
containing_obj=8.950, no_obj=1.434, cls=9.044
Epoch [9/50], Iter [400/418], Loss: total=3.194, reg=18.970,
containing_obj=8.949, no_obj=1.446, cls=8.964
Time for epoch: 189.35 seconds
Updating best test loss: 3.33028

Starting epoch 10 / 50

Learning Rate for this epoch: 0.001

Epoch [10/50], Iter [50/418], Loss: total=3.288, reg=19.716,
containing_obj=9.065, no_obj=1.530, cls=9.146
Epoch [10/50], Iter [100/418], Loss: total=3.198, reg=19.132,
containing_obj=9.143, no_obj=1.517, cls=8.582
Epoch [10/50], Iter [150/418], Loss: total=3.146, reg=18.789,
containing_obj=8.990, no_obj=1.511, cls=8.461
Epoch [10/50], Iter [200/418], Loss: total=3.107, reg=18.629,
containing_obj=8.902, no_obj=1.532, cls=8.219
Epoch [10/50], Iter [250/418], Loss: total=3.110, reg=18.807,

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containing_obj=8.866, no_obj=1.532, cls=8.114
Epoch [10/50], Iter [300/418], Loss: total=3.123, reg=18.764,
containing_obj=8.900, no_obj=1.530, cls=8.282
Epoch [10/50], Iter [350/418], Loss: total=3.118, reg=18.767,
containing_obj=8.911, no_obj=1.544, cls=8.191
Epoch [10/50], Iter [400/418], Loss: total=3.114, reg=18.731,
containing_obj=8.923, no_obj=1.544, cls=8.176
Time for epoch: 191.19 seconds
---Evaluate model on test samples---

100%|      | 4950/4950 [01:24<00:00, 58.70it/s]

---class aeroplane ap 0.3014771097780536---
---class bicycle ap 0.36129658040223733---
---class bird ap 0.2534141665673341---
---class boat ap 0.07343590168929827---
---class bottle ap 0.04227217257341699---
---class bus ap 0.17271955633441674---
---class car ap 0.4169200539376061---
---class cat ap 0.4512988517326965---
---class chair ap 0.15286722434864022---
---class cow ap 0.22241577841893784---
---class diningtable ap 0.028432732316227463---
---class dog ap 0.4464640386148201---
---class horse ap 0.48321168267725856---
---class motorbike ap 0.42621271511382786---
---class person ap 0.3307705571662599---
---class pottedplant ap 0.0510021749463492---
---class sheep ap 0.12924341380581822---
---class sofa ap 0.12966925785902456---
---class train ap 0.3250704121609788---
---class tvmonitor ap 0.2561439547886648---
---map 0.25271691676159336---
9 [np.float64(0.3014771097780536), np.float64(0.36129658040223733),
np.float64(0.2534141665673341), np.float64(0.07343590168929827),
np.float64(0.04227217257341699), np.float64(0.17271955633441674),
np.float64(0.4169200539376061), np.float64(0.4512988517326965),
np.float64(0.15286722434864022), np.float64(0.22241577841893784),
np.float64(0.028432732316227463), np.float64(0.4464640386148201),
np.float64(0.48321168267725856), np.float64(0.42621271511382786),
np.float64(0.3307705571662599), np.float64(0.0510021749463492),
np.float64(0.12924341380581822), np.float64(0.12966925785902456),
np.float64(0.3250704121609788), np.float64(0.2561439547886648)]
Updating best test loss: 3.16849

Starting epoch 11 / 50
Learning Rate for this epoch: 0.001
Epoch [11/50], Iter [50/418], Loss: total=3.174, reg=18.985,

```

containing_obj=9.228, no_obj=1.556, cls=8.318
Epoch [11/50], Iter [100/418], Loss: total=3.045, reg=18.191,
containing_obj=8.792, no_obj=1.574, cls=7.985
Epoch [11/50], Iter [150/418], Loss: total=3.060, reg=18.325,
containing_obj=8.991, no_obj=1.550, cls=7.849
Epoch [11/50], Iter [200/418], Loss: total=2.984, reg=17.839,
containing_obj=8.807, no_obj=1.574, cls=7.588
Epoch [11/50], Iter [250/418], Loss: total=2.960, reg=17.718,
containing_obj=8.764, no_obj=1.581, cls=7.454
Epoch [11/50], Iter [300/418], Loss: total=2.970, reg=17.833,
containing_obj=8.814, no_obj=1.592, cls=7.403
Epoch [11/50], Iter [350/418], Loss: total=2.975, reg=17.939,
containing_obj=8.804, no_obj=1.603, cls=7.349
Epoch [11/50], Iter [400/418], Loss: total=2.995, reg=18.053,
containing_obj=8.849, no_obj=1.605, cls=7.436
Time for epoch: 252.97 seconds
Updating best test loss: 3.08088

Starting epoch 12 / 50
Learning Rate for this epoch: 0.001
Epoch [12/50], Iter [50/418], Loss: total=3.048, reg=18.450,
containing_obj=9.083, no_obj=1.566, cls=7.474
Epoch [12/50], Iter [100/418], Loss: total=2.978, reg=17.995,
containing_obj=8.908, no_obj=1.614, cls=7.219
Epoch [12/50], Iter [150/418], Loss: total=2.920, reg=17.630,
containing_obj=8.782, no_obj=1.628, cls=7.001
Epoch [12/50], Iter [200/418], Loss: total=2.930, reg=17.771,
containing_obj=8.840, no_obj=1.615, cls=6.928
Epoch [12/50], Iter [250/418], Loss: total=2.895, reg=17.597,
containing_obj=8.728, no_obj=1.642, cls=6.768
Epoch [12/50], Iter [300/418], Loss: total=2.905, reg=17.629,
containing_obj=8.778, no_obj=1.629, cls=6.825
Epoch [12/50], Iter [350/418], Loss: total=2.921, reg=17.638,
containing_obj=8.867, no_obj=1.643, cls=6.900
Epoch [12/50], Iter [400/418], Loss: total=2.906, reg=17.612,
containing_obj=8.820, no_obj=1.646, cls=6.790
Time for epoch: 192.05 seconds
Updating best test loss: 3.03889

Starting epoch 13 / 50
Learning Rate for this epoch: 0.001
Epoch [13/50], Iter [50/418], Loss: total=2.817, reg=17.247,
containing_obj=8.435, no_obj=1.685, cls=6.432
Epoch [13/50], Iter [100/418], Loss: total=2.859, reg=17.450,
containing_obj=8.797, no_obj=1.738, cls=6.323
Epoch [13/50], Iter [150/418], Loss: total=2.882, reg=17.660,

containing_obj=8.911, no_obj=1.724, cls=6.294
Epoch [13/50], Iter [200/418], Loss: total=2.839, reg=17.283,
containing_obj=8.760, no_obj=1.741, cls=6.280
Epoch [13/50], Iter [250/418], Loss: total=2.867, reg=17.441,
containing_obj=8.827, no_obj=1.726, cls=6.408
Epoch [13/50], Iter [300/418], Loss: total=2.847, reg=17.302,
containing_obj=8.801, no_obj=1.712, cls=6.345
Epoch [13/50], Iter [350/418], Loss: total=2.836, reg=17.248,
containing_obj=8.799, no_obj=1.711, cls=6.276
Epoch [13/50], Iter [400/418], Loss: total=2.839, reg=17.189,
containing_obj=8.835, no_obj=1.706, cls=6.339
Time for epoch: 190.47 seconds
Updating best test loss: 2.96374

Starting epoch 14 / 50

Learning Rate for this epoch: 0.001

Epoch [14/50], Iter [50/418], Loss: total=2.850, reg=17.577,
containing_obj=9.186, no_obj=1.697, cls=5.741
Epoch [14/50], Iter [100/418], Loss: total=2.795, reg=17.112,
containing_obj=8.833, no_obj=1.769, cls=5.829
Epoch [14/50], Iter [150/418], Loss: total=2.790, reg=17.023,
containing_obj=8.907, no_obj=1.733, cls=5.819
Epoch [14/50], Iter [200/418], Loss: total=2.750, reg=16.665,
containing_obj=8.788, no_obj=1.734, cls=5.810
Epoch [14/50], Iter [250/418], Loss: total=2.747, reg=16.659,
containing_obj=8.741, no_obj=1.748, cls=5.821
Epoch [14/50], Iter [300/418], Loss: total=2.767, reg=16.879,
containing_obj=8.752, no_obj=1.758, cls=5.817
Epoch [14/50], Iter [350/418], Loss: total=2.750, reg=16.773,
containing_obj=8.722, no_obj=1.758, cls=5.746
Epoch [14/50], Iter [400/418], Loss: total=2.746, reg=16.795,
containing_obj=8.683, no_obj=1.762, cls=5.712
Time for epoch: 190.79 seconds
Updating best test loss: 2.91273

Starting epoch 15 / 50

Learning Rate for this epoch: 0.001

Epoch [15/50], Iter [50/418], Loss: total=2.674, reg=16.566,
containing_obj=8.431, no_obj=1.829, cls=5.268
Epoch [15/50], Iter [100/418], Loss: total=2.644, reg=16.099,
containing_obj=8.453, no_obj=1.807, cls=5.375
Epoch [15/50], Iter [150/418], Loss: total=2.650, reg=16.272,
containing_obj=8.411, no_obj=1.802, cls=5.316
Epoch [15/50], Iter [200/418], Loss: total=2.642, reg=16.155,
containing_obj=8.385, no_obj=1.814, cls=5.348
Epoch [15/50], Iter [250/418], Loss: total=2.625, reg=15.985,

```

containing_obj=8.414, no_obj=1.791, cls=5.305
Epoch [15/50], Iter [300/418], Loss: total=2.644, reg=16.155,
containing_obj=8.484, no_obj=1.806, cls=5.277
Epoch [15/50], Iter [350/418], Loss: total=2.659, reg=16.281,
containing_obj=8.514, no_obj=1.801, cls=5.314
Epoch [15/50], Iter [400/418], Loss: total=2.675, reg=16.323,
containing_obj=8.607, no_obj=1.801, cls=5.375
Time for epoch: 190.59 seconds
---Evaluate model on test samples---

100%|      | 4950/4950 [01:24<00:00, 58.53it/s]

---class aeroplane ap 0.3371538592107005---
---class bicycle ap 0.5628979053608303---
---class bird ap 0.26290594121985933---
---class boat ap 0.10352678039158936---
---class bottle ap 0.0912819074928283---
---class bus ap 0.4395032201443323---
---class car ap 0.5177623888302951---
---class cat ap 0.6229717100392994---
---class chair ap 0.18520246852414074---
---class cow ap 0.3323419345483797---
---class diningtable ap 0.23041802301531586---
---class dog ap 0.6099485999051485---
---class horse ap 0.574582701008876---
---class motorbike ap 0.4672229796243358---
---class person ap 0.37390284918553546---
---class pottedplant ap 0.08567068261019709---
---class sheep ap 0.29385692353314485---
---class sofa ap 0.3640430523902115---
---class train ap 0.5569246156871104---
---class tvmonitor ap 0.3572601322150919---
---map 0.3684689337468612---
14 [np.float64(0.3371538592107005), np.float64(0.5628979053608303),
np.float64(0.26290594121985933), np.float64(0.10352678039158936),
np.float64(0.0912819074928283), np.float64(0.4395032201443323),
np.float64(0.5177623888302951), np.float64(0.6229717100392994),
np.float64(0.18520246852414074), np.float64(0.3323419345483797),
np.float64(0.23041802301531586), np.float64(0.6099485999051485),
np.float64(0.574582701008876), np.float64(0.4672229796243358),
np.float64(0.37390284918553546), np.float64(0.08567068261019709),
np.float64(0.29385692353314485), np.float64(0.3640430523902115),
np.float64(0.5569246156871104), np.float64(0.3572601322150919)]
Updating best test loss: 2.89029

Starting epoch 16 / 50
Learning Rate for this epoch: 0.001
Epoch [16/50], Iter [50/418], Loss: total=2.770, reg=17.904,

```

containing_obj=8.783, no_obj=1.905, cls=4.646
Epoch [16/50], Iter [100/418], Loss: total=2.613, reg=16.222,
containing_obj=8.345, no_obj=1.907, cls=4.881
Epoch [16/50], Iter [150/418], Loss: total=2.569, reg=15.823,
containing_obj=8.332, no_obj=1.889, cls=4.780
Epoch [16/50], Iter [200/418], Loss: total=2.604, reg=16.046,
containing_obj=8.441, no_obj=1.884, cls=4.873
Epoch [16/50], Iter [250/418], Loss: total=2.575, reg=15.790,
containing_obj=8.421, no_obj=1.885, cls=4.801
Epoch [16/50], Iter [300/418], Loss: total=2.604, reg=15.958,
containing_obj=8.519, no_obj=1.873, cls=4.899
Epoch [16/50], Iter [350/418], Loss: total=2.598, reg=15.915,
containing_obj=8.483, no_obj=1.886, cls=4.891
Epoch [16/50], Iter [400/418], Loss: total=2.600, reg=15.938,
containing_obj=8.450, no_obj=1.886, cls=4.932
Time for epoch: 191.11 seconds
Updating best test loss: 2.83083

Starting epoch 17 / 50
Learning Rate for this epoch: 0.001
Epoch [17/50], Iter [50/418], Loss: total=2.429, reg=14.680,
containing_obj=8.276, no_obj=1.861, cls=4.327
Epoch [17/50], Iter [100/418], Loss: total=2.417, reg=14.599,
containing_obj=8.149, no_obj=1.949, cls=4.306
Epoch [17/50], Iter [150/418], Loss: total=2.437, reg=14.701,
containing_obj=8.141, no_obj=1.933, cls=4.466
Epoch [17/50], Iter [200/418], Loss: total=2.454, reg=14.854,
containing_obj=8.245, no_obj=1.920, cls=4.434
Epoch [17/50], Iter [250/418], Loss: total=2.520, reg=15.379,
containing_obj=8.385, no_obj=1.919, cls=4.558
Epoch [17/50], Iter [300/418], Loss: total=2.533, reg=15.414,
containing_obj=8.382, no_obj=1.943, cls=4.657
Epoch [17/50], Iter [350/418], Loss: total=2.519, reg=15.300,
containing_obj=8.331, no_obj=1.943, cls=4.651
Epoch [17/50], Iter [400/418], Loss: total=2.527, reg=15.313,
containing_obj=8.373, no_obj=1.945, cls=4.688
Time for epoch: 199.02 seconds
Updating best test loss: 2.81342

Starting epoch 18 / 50
Learning Rate for this epoch: 0.001
Epoch [18/50], Iter [50/418], Loss: total=2.329, reg=14.267,
containing_obj=7.903, no_obj=1.976, cls=3.802
Epoch [18/50], Iter [100/418], Loss: total=2.469, reg=15.219,
containing_obj=8.385, no_obj=1.945, cls=4.083
Epoch [18/50], Iter [150/418], Loss: total=2.518, reg=15.510,

containing_obj=8.422, no_obj=1.924, cls=4.365
Epoch [18/50], Iter [200/418], Loss: total=2.456, reg=14.980,
containing_obj=8.267, no_obj=1.936, cls=4.293
Epoch [18/50], Iter [250/418], Loss: total=2.515, reg=15.384,
containing_obj=8.389, no_obj=1.932, cls=4.470
Epoch [18/50], Iter [300/418], Loss: total=2.525, reg=15.436,
containing_obj=8.372, no_obj=1.938, cls=4.553
Epoch [18/50], Iter [350/418], Loss: total=2.513, reg=15.395,
containing_obj=8.336, no_obj=1.928, cls=4.503
Epoch [18/50], Iter [400/418], Loss: total=2.521, reg=15.427,
containing_obj=8.343, no_obj=1.934, cls=4.544
Time for epoch: 190.77 seconds
Updating best test loss: 2.79451

Starting epoch 19 / 50

Learning Rate for this epoch: 0.001

Epoch [19/50], Iter [50/418], Loss: total=2.429, reg=15.060,
containing_obj=7.796, no_obj=1.994, cls=4.301
Epoch [19/50], Iter [100/418], Loss: total=2.521, reg=15.690,
containing_obj=8.119, no_obj=1.998, cls=4.442
Epoch [19/50], Iter [150/418], Loss: total=2.487, reg=15.223,
containing_obj=8.217, no_obj=2.002, cls=4.397
Epoch [19/50], Iter [200/418], Loss: total=2.486, reg=15.229,
containing_obj=8.286, no_obj=2.003, cls=4.309
Epoch [19/50], Iter [250/418], Loss: total=2.464, reg=15.007,
containing_obj=8.249, no_obj=2.003, cls=4.314
Epoch [19/50], Iter [300/418], Loss: total=2.470, reg=14.993,
containing_obj=8.243, no_obj=2.007, cls=4.392
Epoch [19/50], Iter [350/418], Loss: total=2.473, reg=15.054,
containing_obj=8.247, no_obj=2.008, cls=4.366
Epoch [19/50], Iter [400/418], Loss: total=2.474, reg=15.058,
containing_obj=8.216, no_obj=2.012, cls=4.398
Time for epoch: 190.75 seconds
Updating best test loss: 2.77720

Starting epoch 20 / 50

Learning Rate for this epoch: 0.001

Epoch [20/50], Iter [50/418], Loss: total=2.424, reg=14.727,
containing_obj=8.064, no_obj=2.081, cls=4.218
Epoch [20/50], Iter [100/418], Loss: total=2.447, reg=15.083,
containing_obj=8.211, no_obj=2.074, cls=3.996
Epoch [20/50], Iter [150/418], Loss: total=2.452, reg=15.018,
containing_obj=8.221, no_obj=2.039, cls=4.146
Epoch [20/50], Iter [200/418], Loss: total=2.433, reg=14.893,
containing_obj=8.166, no_obj=2.066, cls=4.071
Epoch [20/50], Iter [250/418], Loss: total=2.408, reg=14.803,

```

containing_obj=8.027, no_obj=2.077, cls=3.990
Epoch [20/50], Iter [300/418], Loss: total=2.387, reg=14.654,
containing_obj=7.971, no_obj=2.066, cls=3.953
Epoch [20/50], Iter [350/418], Loss: total=2.394, reg=14.719,
containing_obj=7.995, no_obj=2.056, cls=3.956
Epoch [20/50], Iter [400/418], Loss: total=2.386, reg=14.631,
containing_obj=8.008, no_obj=2.036, cls=3.959
Time for epoch: 190.89 seconds
---Evaluate model on test samples---

100%|      | 4950/4950 [01:24<00:00, 58.86it/s]

---class aeroplane ap 0.38373665045678396---
---class bicycle ap 0.5357289364138224---
---class bird ap 0.34199662216086213---
---class boat ap 0.19278342433626366---
---class bottle ap 0.13427908521244628---
---class bus ap 0.5095167743278147---
---class car ap 0.6028684117613615---
---class cat ap 0.5345679721047478---
---class chair ap 0.2414396155258963---
---class cow ap 0.3833701882162965---
---class diningtable ap 0.2905840139183931---
---class dog ap 0.6156815830818944---
---class horse ap 0.6450003304307534---
---class motorbike ap 0.47418563162654603---
---class person ap 0.451757499035337---
---class pottedplant ap 0.11735358711965518---
---class sheep ap 0.3620237782139003---
---class sofa ap 0.44845426260646637---
---class train ap 0.5145580299886048---
---class tvmonitor ap 0.3798173970925678---
---map 0.40798518968152064---
19 [np.float64(0.38373665045678396), np.float64(0.5357289364138224),
np.float64(0.34199662216086213), np.float64(0.19278342433626366),
np.float64(0.13427908521244628), np.float64(0.5095167743278147),
np.float64(0.6028684117613615), np.float64(0.5345679721047478),
np.float64(0.2414396155258963), np.float64(0.3833701882162965),
np.float64(0.2905840139183931), np.float64(0.6156815830818944),
np.float64(0.6450003304307534), np.float64(0.47418563162654603),
np.float64(0.451757499035337), np.float64(0.11735358711965518),
np.float64(0.3620237782139003), np.float64(0.44845426260646637),
np.float64(0.5145580299886048), np.float64(0.3798173970925678)]
Updating best test loss: 2.75908

Starting epoch 21 / 50
Learning Rate for this epoch: 0.001
Epoch [21/50], Iter [50/418], Loss: total=2.357, reg=14.120,

```

containing_obj=8.000, no_obj=2.111, cls=4.057
Epoch [21/50], Iter [100/418], Loss: total=2.312, reg=14.050,
containing_obj=7.895, no_obj=2.095, cls=3.707
Epoch [21/50], Iter [150/418], Loss: total=2.357, reg=14.369,
containing_obj=8.012, no_obj=2.089, cls=3.816
Epoch [21/50], Iter [200/418], Loss: total=2.385, reg=14.594,
containing_obj=8.087, no_obj=2.090, cls=3.849
Epoch [21/50], Iter [250/418], Loss: total=2.413, reg=14.776,
containing_obj=8.150, no_obj=2.076, cls=3.954
Epoch [21/50], Iter [300/418], Loss: total=2.358, reg=14.408,
containing_obj=8.019, no_obj=2.069, cls=3.806
Epoch [21/50], Iter [350/418], Loss: total=2.359, reg=14.358,
containing_obj=8.029, no_obj=2.064, cls=3.855
Epoch [21/50], Iter [400/418], Loss: total=2.365, reg=14.432,
containing_obj=8.042, no_obj=2.052, cls=3.860
Time for epoch: 190.78 seconds
Updating best test loss: 2.72059

Starting epoch 22 / 50
Learning Rate for this epoch: 0.001
Epoch [22/50], Iter [50/418], Loss: total=2.288, reg=13.573,
containing_obj=7.852, no_obj=2.184, cls=3.842
Epoch [22/50], Iter [100/418], Loss: total=2.249, reg=13.306,
containing_obj=7.680, no_obj=2.169, cls=3.838
Epoch [22/50], Iter [150/418], Loss: total=2.264, reg=13.482,
containing_obj=7.735, no_obj=2.119, cls=3.831
Epoch [22/50], Iter [200/418], Loss: total=2.307, reg=13.840,
containing_obj=7.859, no_obj=2.117, cls=3.865
Epoch [22/50], Iter [250/418], Loss: total=2.309, reg=13.931,
containing_obj=7.846, no_obj=2.118, cls=3.811
Epoch [22/50], Iter [300/418], Loss: total=2.321, reg=14.091,
containing_obj=7.918, no_obj=2.101, cls=3.741
Epoch [22/50], Iter [350/418], Loss: total=2.331, reg=14.164,
containing_obj=7.943, no_obj=2.122, cls=3.747
Epoch [22/50], Iter [400/418], Loss: total=2.318, reg=14.093,
containing_obj=7.894, no_obj=2.123, cls=3.703
Time for epoch: 190.82 seconds
Updating best test loss: 2.69874

Starting epoch 23 / 50
Learning Rate for this epoch: 0.001
Epoch [23/50], Iter [50/418], Loss: total=2.218, reg=13.180,
containing_obj=7.995, no_obj=2.025, cls=3.417
Epoch [23/50], Iter [100/418], Loss: total=2.195, reg=12.982,
containing_obj=7.932, no_obj=2.063, cls=3.366
Epoch [23/50], Iter [150/418], Loss: total=2.217, reg=13.213,

containing_obj=7.911, no_obj=2.094, cls=3.385
Epoch [23/50], Iter [200/418], Loss: total=2.246, reg=13.514,
containing_obj=7.883, no_obj=2.077, cls=3.480
Epoch [23/50], Iter [250/418], Loss: total=2.246, reg=13.487,
containing_obj=7.895, no_obj=2.082, cls=3.486
Epoch [23/50], Iter [300/418], Loss: total=2.240, reg=13.457,
containing_obj=7.926, no_obj=2.079, cls=3.423
Epoch [23/50], Iter [350/418], Loss: total=2.236, reg=13.432,
containing_obj=7.864, no_obj=2.095, cls=3.445
Epoch [23/50], Iter [400/418], Loss: total=2.263, reg=13.615,
containing_obj=7.954, no_obj=2.089, cls=3.495
Time for epoch: 191.04 seconds
Updating best test loss: 2.67597

Starting epoch 24 / 50

Learning Rate for this epoch: 0.001

Epoch [24/50], Iter [50/418], Loss: total=2.254, reg=13.890,
containing_obj=7.789, no_obj=2.113, cls=3.259
Epoch [24/50], Iter [100/418], Loss: total=2.282, reg=13.935,
containing_obj=7.935, no_obj=2.138, cls=3.371
Epoch [24/50], Iter [150/418], Loss: total=2.258, reg=13.837,
containing_obj=7.786, no_obj=2.160, cls=3.312
Epoch [24/50], Iter [200/418], Loss: total=2.276, reg=13.911,
containing_obj=7.858, no_obj=2.166, cls=3.375
Epoch [24/50], Iter [250/418], Loss: total=2.275, reg=13.877,
containing_obj=7.835, no_obj=2.181, cls=3.413
Epoch [24/50], Iter [300/418], Loss: total=2.255, reg=13.686,
containing_obj=7.784, no_obj=2.164, cls=3.425
Epoch [24/50], Iter [350/418], Loss: total=2.263, reg=13.795,
containing_obj=7.765, no_obj=2.144, cls=3.449
Epoch [24/50], Iter [400/418], Loss: total=2.265, reg=13.780,
containing_obj=7.755, no_obj=2.153, cls=3.486
Time for epoch: 190.98 seconds

Starting epoch 25 / 50

Learning Rate for this epoch: 0.001

Epoch [25/50], Iter [50/418], Loss: total=2.105, reg=12.610,
containing_obj=7.589, no_obj=2.092, cls=2.974
Epoch [25/50], Iter [100/418], Loss: total=2.186, reg=13.165,
containing_obj=7.767, no_obj=2.014, cls=3.285
Epoch [25/50], Iter [150/418], Loss: total=2.222, reg=13.566,
containing_obj=7.755, no_obj=2.069, cls=3.277
Epoch [25/50], Iter [200/418], Loss: total=2.171, reg=13.150,
containing_obj=7.631, no_obj=2.109, cls=3.167
Epoch [25/50], Iter [250/418], Loss: total=2.188, reg=13.262,
containing_obj=7.707, no_obj=2.112, cls=3.179

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Epoch [25/50], Iter [300/418], Loss: total=2.194, reg=13.297,
containing_obj=7.767, no_obj=2.121, cls=3.142
Epoch [25/50], Iter [350/418], Loss: total=2.218, reg=13.396,
containing_obj=7.815, no_obj=2.140, cls=3.267
Epoch [25/50], Iter [400/418], Loss: total=2.245, reg=13.546,
containing_obj=7.904, no_obj=2.145, cls=3.343
Time for epoch: 190.87 seconds
---Evaluate model on test samples---
100%|          | 4950/4950 [01:24<00:00, 58.83it/s]

---class aeroplane ap 0.4217317093839185---
---class bicycle ap 0.5373684076566265---
---class bird ap 0.37991870511746473---
---class boat ap 0.22659955885333563---
---class bottle ap 0.16729627391709168---
---class bus ap 0.5613285817030088---
---class car ap 0.610380774473679---
---class cat ap 0.6397721519817997---
---class chair ap 0.2741128403584525---
---class cow ap 0.45511647765176577---
---class diningtable ap 0.25192022418775467---
---class dog ap 0.5726688721837077---
---class horse ap 0.5714826852390325---
---class motorbike ap 0.45233401065535206---
---class person ap 0.4636605870602829---
---class pottedplant ap 0.16427204096028078---
---class sheep ap 0.3764870121645443---
---class sofa ap 0.4753121219304392---
---class train ap 0.5796648386087666---
---class tvmonitor ap 0.4579704628781178---
---map 0.431969916848271---
24 [np.float64(0.4217317093839185), np.float64(0.5373684076566265),
np.float64(0.37991870511746473), np.float64(0.22659955885333563),
np.float64(0.16729627391709168), np.float64(0.5613285817030088),
np.float64(0.610380774473679), np.float64(0.6397721519817997),
np.float64(0.2741128403584525), np.float64(0.45511647765176577),
np.float64(0.25192022418775467), np.float64(0.5726688721837077),
np.float64(0.5714826852390325), np.float64(0.45233401065535206),
np.float64(0.4636605870602829), np.float64(0.16427204096028078),
np.float64(0.3764870121645443), np.float64(0.4753121219304392),
np.float64(0.5796648386087666), np.float64(0.4579704628781178)]

Starting epoch 26 / 50
Learning Rate for this epoch: 0.001
Epoch [26/50], Iter [50/418], Loss: total=2.001, reg=11.962,
containing_obj=7.151, no_obj=2.124, cls=2.776
Epoch [26/50], Iter [100/418], Loss: total=2.058, reg=12.375,

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containing_obj=7.211, no_obj=2.169, cls=2.938
Epoch [26/50], Iter [150/418], Loss: total=2.111, reg=12.609,
containing_obj=7.446, no_obj=2.145, cls=3.133
Epoch [26/50], Iter [200/418], Loss: total=2.136, reg=12.798,
containing_obj=7.577, no_obj=2.179, cls=3.074
Epoch [26/50], Iter [250/418], Loss: total=2.186, reg=13.157,
containing_obj=7.678, no_obj=2.187, cls=3.205
Epoch [26/50], Iter [300/418], Loss: total=2.154, reg=12.955,
containing_obj=7.551, no_obj=2.178, cls=3.162
Epoch [26/50], Iter [350/418], Loss: total=2.168, reg=13.076,
containing_obj=7.628, no_obj=2.174, cls=3.134
Epoch [26/50], Iter [400/418], Loss: total=2.181, reg=13.106,
containing_obj=7.709, no_obj=2.177, cls=3.178
Time for epoch: 190.66 seconds
Updating best test loss: 2.62829

Starting epoch 27 / 50

Learning Rate for this epoch: 0.001
Epoch [27/50], Iter [50/418], Loss: total=2.084, reg=12.805,
containing_obj=7.203, no_obj=2.212, cls=2.789
Epoch [27/50], Iter [100/418], Loss: total=2.168, reg=13.316,
containing_obj=7.534, no_obj=2.238, cls=2.928
Epoch [27/50], Iter [150/418], Loss: total=2.167, reg=13.088,
containing_obj=7.582, no_obj=2.223, cls=3.111
Epoch [27/50], Iter [200/418], Loss: total=2.156, reg=13.018,
containing_obj=7.544, no_obj=2.205, cls=3.105
Epoch [27/50], Iter [250/418], Loss: total=2.133, reg=12.843,
containing_obj=7.529, no_obj=2.180, cls=3.040
Epoch [27/50], Iter [300/418], Loss: total=2.135, reg=12.803,
containing_obj=7.593, no_obj=2.188, cls=3.035
Epoch [27/50], Iter [350/418], Loss: total=2.127, reg=12.730,
containing_obj=7.571, no_obj=2.193, cls=3.030
Epoch [27/50], Iter [400/418], Loss: total=2.131, reg=12.815,
containing_obj=7.552, no_obj=2.195, cls=3.015
Time for epoch: 190.90 seconds

Starting epoch 28 / 50

Learning Rate for this epoch: 0.001
Epoch [28/50], Iter [50/418], Loss: total=2.050, reg=12.562,
containing_obj=7.130, no_obj=2.208, cls=2.705
Epoch [28/50], Iter [100/418], Loss: total=2.106, reg=12.980,
containing_obj=7.221, no_obj=2.179, cls=2.894
Epoch [28/50], Iter [150/418], Loss: total=2.070, reg=12.677,
containing_obj=7.247, no_obj=2.187, cls=2.731
Epoch [28/50], Iter [200/418], Loss: total=2.089, reg=12.758,
containing_obj=7.383, no_obj=2.192, cls=2.738

Epoch [28/50], Iter [250/418], Loss: total=2.079, reg=12.604,
containing_obj=7.401, no_obj=2.194, cls=2.744
Epoch [28/50], Iter [300/418], Loss: total=2.086, reg=12.590,
containing_obj=7.432, no_obj=2.196, cls=2.808
Epoch [28/50], Iter [350/418], Loss: total=2.097, reg=12.699,
containing_obj=7.482, no_obj=2.182, cls=2.806
Epoch [28/50], Iter [400/418], Loss: total=2.112, reg=12.778,
containing_obj=7.541, no_obj=2.184, cls=2.844
Time for epoch: 190.95 seconds

Starting epoch 29 / 50

Learning Rate for this epoch: 0.001

Epoch [29/50], Iter [50/418], Loss: total=2.287, reg=13.776,
containing_obj=8.457, no_obj=2.191, cls=3.024
Epoch [29/50], Iter [100/418], Loss: total=2.221, reg=13.383,
containing_obj=7.880, no_obj=2.260, cls=3.132
Epoch [29/50], Iter [150/418], Loss: total=2.167, reg=13.034,
containing_obj=7.769, no_obj=2.264, cls=2.941
Epoch [29/50], Iter [200/418], Loss: total=2.115, reg=12.642,
containing_obj=7.635, no_obj=2.268, cls=2.838
Epoch [29/50], Iter [250/418], Loss: total=2.101, reg=12.533,
containing_obj=7.586, no_obj=2.236, cls=2.860
Epoch [29/50], Iter [300/418], Loss: total=2.108, reg=12.598,
containing_obj=7.590, no_obj=2.218, cls=2.885
Epoch [29/50], Iter [350/418], Loss: total=2.118, reg=12.602,
containing_obj=7.666, no_obj=2.223, cls=2.922
Epoch [29/50], Iter [400/418], Loss: total=2.116, reg=12.589,
containing_obj=7.614, no_obj=2.223, cls=2.966
Time for epoch: 191.11 seconds

Starting epoch 30 / 50

Learning Rate for this epoch: 0.001

Epoch [30/50], Iter [50/418], Loss: total=2.113, reg=12.897,
containing_obj=7.507, no_obj=2.115, cls=2.832
Epoch [30/50], Iter [100/418], Loss: total=2.076, reg=12.422,
containing_obj=7.490, no_obj=2.250, cls=2.752
Epoch [30/50], Iter [150/418], Loss: total=2.039, reg=12.201,
containing_obj=7.300, no_obj=2.209, cls=2.752
Epoch [30/50], Iter [200/418], Loss: total=2.055, reg=12.289,
containing_obj=7.384, no_obj=2.240, cls=2.744
Epoch [30/50], Iter [250/418], Loss: total=2.035, reg=12.116,
containing_obj=7.402, no_obj=2.245, cls=2.659
Epoch [30/50], Iter [300/418], Loss: total=2.043, reg=12.179,
containing_obj=7.426, no_obj=2.226, cls=2.685
Epoch [30/50], Iter [350/418], Loss: total=2.043, reg=12.210,
containing_obj=7.413, no_obj=2.228, cls=2.671

```

Epoch [30/50], Iter [400/418], Loss: total=2.060, reg=12.361,
containing_obj=7.448, no_obj=2.220, cls=2.693
Time for epoch: 190.96 seconds
---Evaluate model on test samples---

100%|      | 4950/4950 [01:24<00:00, 58.74it/s]

---class aeroplane ap 0.42918222030881---
---class bicycle ap 0.6205642510255927---
---class bird ap 0.4471715379161858---
---class boat ap 0.2587868237142986---
---class bottle ap 0.17507680789311492---
---class bus ap 0.5893322340909688---
---class car ap 0.6497255806086264---
---class cat ap 0.6738586006640961---
---class chair ap 0.25414065272068614---
---class cow ap 0.5156951850644542---
---class diningtable ap 0.34566773606517404---
---class dog ap 0.6403550895129018---
---class horse ap 0.6654910046936673---
---class motorbike ap 0.5576233402742758---
---class person ap 0.4872254592979751---
---class pottedplant ap 0.1679885050521627---
---class sheep ap 0.49208715898819116---
---class sofa ap 0.4785689345066969---
---class train ap 0.6550972961108075---
---class tvmonitor ap 0.47834335800505523---
---map 0.47909908882568714---
29 [np.float64(0.42918222030881), np.float64(0.6205642510255927),
np.float64(0.4471715379161858), np.float64(0.2587868237142986),
np.float64(0.17507680789311492), np.float64(0.5893322340909688),
np.float64(0.6497255806086264), np.float64(0.6738586006640961),
np.float64(0.25414065272068614), np.float64(0.5156951850644542),
np.float64(0.34566773606517404), np.float64(0.6403550895129018),
np.float64(0.6654910046936673), np.float64(0.5576233402742758),
np.float64(0.4872254592979751), np.float64(0.1679885050521627),
np.float64(0.49208715898819116), np.float64(0.4785689345066969),
np.float64(0.6550972961108075), np.float64(0.47834335800505523)]
Updating best test loss: 2.58625

Starting epoch 31 / 50
Learning Rate for this epoch: 0.0001
Epoch [31/50], Iter [50/418], Loss: total=2.063, reg=12.112,
containing_obj=7.667, no_obj=2.180, cls=2.797
Epoch [31/50], Iter [100/418], Loss: total=1.983, reg=11.485,
containing_obj=7.393, no_obj=2.208, cls=2.707
Epoch [31/50], Iter [150/418], Loss: total=1.971, reg=11.376,
containing_obj=7.449, no_obj=2.258, cls=2.563

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Epoch [31/50], Iter [200/418], Loss: total=1.968, reg=11.454,
containing_obj=7.375, no_obj=2.265, cls=2.523
Epoch [31/50], Iter [250/418], Loss: total=1.996, reg=11.682,
containing_obj=7.447, no_obj=2.263, cls=2.555
Epoch [31/50], Iter [300/418], Loss: total=1.989, reg=11.679,
containing_obj=7.427, no_obj=2.259, cls=2.508
Epoch [31/50], Iter [350/418], Loss: total=1.985, reg=11.638,
containing_obj=7.408, no_obj=2.263, cls=2.511
Epoch [31/50], Iter [400/418], Loss: total=2.000, reg=11.743,
containing_obj=7.471, no_obj=2.256, cls=2.535
Time for epoch: 190.89 seconds
Updating best test loss: 2.54931

Starting epoch 32 / 50

Learning Rate for this epoch: 0.0001

Epoch [32/50], Iter [50/418], Loss: total=2.066, reg=12.112,
containing_obj=7.803, no_obj=2.248, cls=2.628
Epoch [32/50], Iter [100/418], Loss: total=2.008, reg=12.027,
containing_obj=7.555, no_obj=2.211, cls=2.302
Epoch [32/50], Iter [150/418], Loss: total=2.013, reg=12.022,
containing_obj=7.517, no_obj=2.226, cls=2.386
Epoch [32/50], Iter [200/418], Loss: total=1.999, reg=11.934,
containing_obj=7.464, no_obj=2.247, cls=2.347
Epoch [32/50], Iter [250/418], Loss: total=1.982, reg=11.806,
containing_obj=7.366, no_obj=2.264, cls=2.347
Epoch [32/50], Iter [300/418], Loss: total=1.982, reg=11.753,
containing_obj=7.344, no_obj=2.264, cls=2.423
Epoch [32/50], Iter [350/418], Loss: total=1.974, reg=11.719,
containing_obj=7.311, no_obj=2.268, cls=2.394
Epoch [32/50], Iter [400/418], Loss: total=1.961, reg=11.601,
containing_obj=7.267, no_obj=2.270, cls=2.389
Time for epoch: 190.79 seconds
Updating best test loss: 2.54848

Starting epoch 33 / 50

Learning Rate for this epoch: 0.0001

Epoch [33/50], Iter [50/418], Loss: total=1.868, reg=11.145,
containing_obj=6.955, no_obj=2.366, cls=1.947
Epoch [33/50], Iter [100/418], Loss: total=1.888, reg=11.212,
containing_obj=7.148, no_obj=2.324, cls=1.971
Epoch [33/50], Iter [150/418], Loss: total=1.950, reg=11.681,
containing_obj=7.321, no_obj=2.316, cls=2.085
Epoch [33/50], Iter [200/418], Loss: total=1.949, reg=11.507,
containing_obj=7.339, no_obj=2.311, cls=2.229
Epoch [33/50], Iter [250/418], Loss: total=1.955, reg=11.511,
containing_obj=7.372, no_obj=2.321, cls=2.256

Epoch [33/50], Iter [300/418], Loss: total=1.937, reg=11.420,
containing_obj=7.302, no_obj=2.308, cls=2.220
Epoch [33/50], Iter [350/418], Loss: total=1.933, reg=11.363,
containing_obj=7.273, no_obj=2.325, cls=2.231
Epoch [33/50], Iter [400/418], Loss: total=1.928, reg=11.318,
containing_obj=7.257, no_obj=2.335, cls=2.222
Time for epoch: 190.30 seconds
Updating best test loss: 2.54219

Starting epoch 34 / 50

Learning Rate for this epoch: 0.0001

Epoch [34/50], Iter [50/418], Loss: total=1.997, reg=11.818,
containing_obj=7.487, no_obj=2.295, cls=2.368
Epoch [34/50], Iter [100/418], Loss: total=1.942, reg=11.396,
containing_obj=7.316, no_obj=2.303, cls=2.289
Epoch [34/50], Iter [150/418], Loss: total=1.960, reg=11.508,
containing_obj=7.266, no_obj=2.299, cls=2.446
Epoch [34/50], Iter [200/418], Loss: total=1.940, reg=11.313,
containing_obj=7.187, no_obj=2.302, cls=2.480
Epoch [34/50], Iter [250/418], Loss: total=1.920, reg=11.189,
containing_obj=7.161, no_obj=2.288, cls=2.399
Epoch [34/50], Iter [300/418], Loss: total=1.941, reg=11.327,
containing_obj=7.175, no_obj=2.299, cls=2.485
Epoch [34/50], Iter [350/418], Loss: total=1.942, reg=11.381,
containing_obj=7.182, no_obj=2.293, cls=2.446
Epoch [34/50], Iter [400/418], Loss: total=1.947, reg=11.454,
containing_obj=7.214, no_obj=2.283, cls=2.416
Time for epoch: 191.06 seconds
Updating best test loss: 2.54180

Starting epoch 35 / 50

Learning Rate for this epoch: 0.0001

Epoch [35/50], Iter [50/418], Loss: total=1.969, reg=11.617,
containing_obj=7.367, no_obj=2.388, cls=2.256
Epoch [35/50], Iter [100/418], Loss: total=1.921, reg=11.220,
containing_obj=7.050, no_obj=2.385, cls=2.393
Epoch [35/50], Iter [150/418], Loss: total=1.902, reg=11.146,
containing_obj=7.080, no_obj=2.379, cls=2.217
Epoch [35/50], Iter [200/418], Loss: total=1.914, reg=11.304,
containing_obj=7.107, no_obj=2.364, cls=2.197
Epoch [35/50], Iter [250/418], Loss: total=1.918, reg=11.215,
containing_obj=7.164, no_obj=2.356, cls=2.286
Epoch [35/50], Iter [300/418], Loss: total=1.935, reg=11.416,
containing_obj=7.181, no_obj=2.336, cls=2.291
Epoch [35/50], Iter [350/418], Loss: total=1.933, reg=11.392,
containing_obj=7.176, no_obj=2.324, cls=2.305

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Epoch [35/50], Iter [400/418], Loss: total=1.947, reg=11.494,
containing_obj=7.198, no_obj=2.320, cls=2.351
Time for epoch: 191.29 seconds
---Evaluate model on test samples---

100%|      | 4950/4950 [01:24<00:00, 58.50it/s]

---class aeroplane ap 0.5172145679495267---
---class bicycle ap 0.6211838749383044---
---class bird ap 0.49730149800297263---
---class boat ap 0.2647059985070107---
---class bottle ap 0.17910886137230508---
---class bus ap 0.639861653167359---
---class car ap 0.6761908567119271---
---class cat ap 0.6722196924158539---
---class chair ap 0.2762209804659096---
---class cow ap 0.5511939429792145---
---class diningtable ap 0.34707210287102497---
---class dog ap 0.6434267241183255---
---class horse ap 0.6902102440972934---
---class motorbike ap 0.5779051637612193---
---class person ap 0.5106499239492659---
---class pottedplant ap 0.20401319076236082---
---class sheep ap 0.5072537015575169---
---class sofa ap 0.4995483771764463---
---class train ap 0.6482177512889804---
---class tvmonitor ap 0.5031377423173237---
---map 0.5013318424205069---
34 [np.float64(0.5172145679495267), np.float64(0.6211838749383044),
np.float64(0.49730149800297263), np.float64(0.2647059985070107),
np.float64(0.17910886137230508), np.float64(0.639861653167359),
np.float64(0.6761908567119271), np.float64(0.6722196924158539),
np.float64(0.2762209804659096), np.float64(0.5511939429792145),
np.float64(0.34707210287102497), np.float64(0.6434267241183255),
np.float64(0.6902102440972934), np.float64(0.5779051637612193),
np.float64(0.5106499239492659), np.float64(0.20401319076236082),
np.float64(0.5072537015575169), np.float64(0.4995483771764463),
np.float64(0.6482177512889804), np.float64(0.5031377423173237)]
Updating best test loss: 2.53296

Starting epoch 36 / 50
Learning Rate for this epoch: 0.0001
Epoch [36/50], Iter [50/418], Loss: total=1.898, reg=11.031,
containing_obj=6.953, no_obj=2.371, cls=2.416
Epoch [36/50], Iter [100/418], Loss: total=1.891, reg=10.908,
containing_obj=7.069, no_obj=2.332, cls=2.379
Epoch [36/50], Iter [150/418], Loss: total=1.883, reg=11.039,
containing_obj=7.037, no_obj=2.321, cls=2.202

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Epoch [36/50], Iter [200/418], Loss: total=1.890, reg=11.023,
containing_obj=7.082, no_obj=2.328, cls=2.250
Epoch [36/50], Iter [250/418], Loss: total=1.895, reg=11.083,
containing_obj=7.095, no_obj=2.317, cls=2.240
Epoch [36/50], Iter [300/418], Loss: total=1.906, reg=11.125,
containing_obj=7.178, no_obj=2.309, cls=2.263
Epoch [36/50], Iter [350/418], Loss: total=1.907, reg=11.067,
containing_obj=7.219, no_obj=2.310, cls=2.283
Epoch [36/50], Iter [400/418], Loss: total=1.899, reg=11.058,
containing_obj=7.153, no_obj=2.319, cls=2.258
Time for epoch: 192.34 seconds
Updating best test loss: 2.52952

Starting epoch 37 / 50

Learning Rate for this epoch: 0.0001

Epoch [37/50], Iter [50/418], Loss: total=1.857, reg=11.008,
containing_obj=7.126, no_obj=2.271, cls=1.878
Epoch [37/50], Iter [100/418], Loss: total=1.912, reg=11.439,
containing_obj=7.188, no_obj=2.244, cls=2.073
Epoch [37/50], Iter [150/418], Loss: total=1.904, reg=11.399,
containing_obj=7.140, no_obj=2.204, cls=2.101
Epoch [37/50], Iter [200/418], Loss: total=1.932, reg=11.557,
containing_obj=7.171, no_obj=2.237, cls=2.215
Epoch [37/50], Iter [250/418], Loss: total=1.929, reg=11.470,
containing_obj=7.219, no_obj=2.237, cls=2.227
Epoch [37/50], Iter [300/418], Loss: total=1.933, reg=11.585,
containing_obj=7.158, no_obj=2.251, cls=2.204
Epoch [37/50], Iter [350/418], Loss: total=1.934, reg=11.565,
containing_obj=7.167, no_obj=2.251, cls=2.229
Epoch [37/50], Iter [400/418], Loss: total=1.930, reg=11.505,
containing_obj=7.175, no_obj=2.250, cls=2.226
Time for epoch: 192.42 seconds

Starting epoch 38 / 50

Learning Rate for this epoch: 0.0001

Epoch [38/50], Iter [50/418], Loss: total=1.819, reg=10.444,
containing_obj=6.812, no_obj=2.288, cls=2.284
Epoch [38/50], Iter [100/418], Loss: total=1.849, reg=10.823,
containing_obj=6.829, no_obj=2.304, cls=2.228
Epoch [38/50], Iter [150/418], Loss: total=1.896, reg=11.147,
containing_obj=7.074, no_obj=2.269, cls=2.266
Epoch [38/50], Iter [200/418], Loss: total=1.892, reg=11.107,
containing_obj=7.093, no_obj=2.281, cls=2.219
Epoch [38/50], Iter [250/418], Loss: total=1.903, reg=11.203,
containing_obj=7.113, no_obj=2.286, cls=2.235
Epoch [38/50], Iter [300/418], Loss: total=1.902, reg=11.165,

containing_obj=7.127, no_obj=2.294, cls=2.243
Epoch [38/50], Iter [350/418], Loss: total=1.914, reg=11.215,
containing_obj=7.182, no_obj=2.285, cls=2.284
Epoch [38/50], Iter [400/418], Loss: total=1.909, reg=11.225,
containing_obj=7.160, no_obj=2.281, cls=2.241
Time for epoch: 192.56 seconds

Starting epoch 39 / 50

Learning Rate for this epoch: 0.0001

Epoch [39/50], Iter [50/418], Loss: total=1.773, reg=10.298,
containing_obj=6.682, no_obj=2.411, cls=1.881
Epoch [39/50], Iter [100/418], Loss: total=1.900, reg=11.147,
containing_obj=7.102, no_obj=2.312, cls=2.235
Epoch [39/50], Iter [150/418], Loss: total=1.884, reg=11.058,
containing_obj=7.106, no_obj=2.314, cls=2.131
Epoch [39/50], Iter [200/418], Loss: total=1.874, reg=11.011,
containing_obj=7.022, no_obj=2.295, cls=2.166
Epoch [39/50], Iter [250/418], Loss: total=1.882, reg=11.002,
containing_obj=7.096, no_obj=2.294, cls=2.195
Epoch [39/50], Iter [300/418], Loss: total=1.876, reg=11.027,
containing_obj=7.010, no_obj=2.300, cls=2.169
Epoch [39/50], Iter [350/418], Loss: total=1.900, reg=11.231,
containing_obj=7.117, no_obj=2.303, cls=2.153
Epoch [39/50], Iter [400/418], Loss: total=1.912, reg=11.346,
containing_obj=7.132, no_obj=2.296, cls=2.165
Time for epoch: 192.37 seconds
Updating best test loss: 2.52028

Starting epoch 40 / 50

Learning Rate for this epoch: 0.0001

Epoch [40/50], Iter [50/418], Loss: total=1.791, reg=10.597,
containing_obj=6.973, no_obj=2.314, cls=1.612
Epoch [40/50], Iter [100/418], Loss: total=1.845, reg=10.999,
containing_obj=6.982, no_obj=2.287, cls=1.871
Epoch [40/50], Iter [150/418], Loss: total=1.873, reg=11.053,
containing_obj=7.082, no_obj=2.284, cls=2.054
Epoch [40/50], Iter [200/418], Loss: total=1.875, reg=11.014,
containing_obj=7.070, no_obj=2.289, cls=2.133
Epoch [40/50], Iter [250/418], Loss: total=1.875, reg=11.014,
containing_obj=7.087, no_obj=2.282, cls=2.118
Epoch [40/50], Iter [300/418], Loss: total=1.888, reg=11.054,
containing_obj=7.158, no_obj=2.295, cls=2.149
Epoch [40/50], Iter [350/418], Loss: total=1.884, reg=11.048,
containing_obj=7.133, no_obj=2.297, cls=2.133
Epoch [40/50], Iter [400/418], Loss: total=1.896, reg=11.156,
containing_obj=7.121, no_obj=2.293, cls=2.177

```

Time for epoch: 192.59 seconds
---Evaluate model on test samples---
100%|      | 4950/4950 [01:24<00:00, 58.54it/s]

---class aeroplane ap 0.5454054394869845---
---class bicycle ap 0.6221017178885075---
---class bird ap 0.5169982257513247---
---class boat ap 0.30084837128500574---
---class bottle ap 0.20398418726967044---
---class bus ap 0.6347290978950576---
---class car ap 0.6779320696243094---
---class cat ap 0.698474195009899---
---class chair ap 0.2879846474477097---
---class cow ap 0.5422101372053073---
---class diningtable ap 0.37498605595537826---
---class dog ap 0.6536593565219815---
---class horse ap 0.6949596783568934---
---class motorbike ap 0.5878443833189635---
---class person ap 0.49499801795753773---
---class pottedplant ap 0.24380271222725589---
---class sheep ap 0.4964193389150033---
---class sofa ap 0.5270604121211506---
---class train ap 0.6609148871902215---
---class tvmonitor ap 0.5087950549610453---
---map 0.5137053993194604---
39 [np.float64(0.5454054394869845), np.float64(0.6221017178885075),
np.float64(0.5169982257513247), np.float64(0.30084837128500574),
np.float64(0.20398418726967044), np.float64(0.6347290978950576),
np.float64(0.6779320696243094), np.float64(0.698474195009899),
np.float64(0.2879846474477097), np.float64(0.5422101372053073),
np.float64(0.37498605595537826), np.float64(0.6536593565219815),
np.float64(0.6949596783568934), np.float64(0.5878443833189635),
np.float64(0.49499801795753773), np.float64(0.24380271222725589),
np.float64(0.4964193389150033), np.float64(0.5270604121211506),
np.float64(0.6609148871902215), np.float64(0.5087950549610453)]
Updating best test loss: 2.51906

Starting epoch 41 / 50
Learning Rate for this epoch: 1e-05
Epoch [41/50], Iter [50/418], Loss: total=1.804, reg=10.647,
containing_obj=6.709, no_obj=2.334, cls=1.963
Epoch [41/50], Iter [100/418], Loss: total=1.823, reg=10.674,
containing_obj=6.895, no_obj=2.286, cls=2.026
Epoch [41/50], Iter [150/418], Loss: total=1.821, reg=10.625,
containing_obj=6.829, no_obj=2.301, cls=2.099
Epoch [41/50], Iter [200/418], Loss: total=1.835, reg=10.809,
containing_obj=6.872, no_obj=2.295, cls=2.041

```

Epoch [41/50], Iter [250/418], Loss: total=1.875, reg=11.026,
containing_obj=7.093, no_obj=2.279, cls=2.104
Epoch [41/50], Iter [300/418], Loss: total=1.858, reg=10.867,
containing_obj=7.016, no_obj=2.277, cls=2.131
Epoch [41/50], Iter [350/418], Loss: total=1.852, reg=10.772,
containing_obj=7.029, no_obj=2.277, cls=2.146
Epoch [41/50], Iter [400/418], Loss: total=1.870, reg=10.891,
containing_obj=7.089, no_obj=2.279, cls=2.184
Time for epoch: 192.65 seconds
Updating best test loss: 2.51815

Starting epoch 42 / 50
Learning Rate for this epoch: 1e-05
Epoch [42/50], Iter [50/418], Loss: total=1.858, reg=11.161,
containing_obj=6.622, no_obj=2.322, cls=2.189
Epoch [42/50], Iter [100/418], Loss: total=1.879, reg=11.171,
containing_obj=6.829, no_obj=2.262, cls=2.283
Epoch [42/50], Iter [150/418], Loss: total=1.869, reg=11.026,
containing_obj=6.951, no_obj=2.270, cls=2.185
Epoch [42/50], Iter [200/418], Loss: total=1.860, reg=10.968,
containing_obj=7.016, no_obj=2.259, cls=2.083
Epoch [42/50], Iter [250/418], Loss: total=1.861, reg=11.005,
containing_obj=7.005, no_obj=2.250, cls=2.071
Epoch [42/50], Iter [300/418], Loss: total=1.868, reg=11.016,
containing_obj=7.076, no_obj=2.253, cls=2.073
Epoch [42/50], Iter [350/418], Loss: total=1.881, reg=11.117,
containing_obj=7.075, no_obj=2.256, cls=2.126
Epoch [42/50], Iter [400/418], Loss: total=1.875, reg=11.045,
containing_obj=7.087, no_obj=2.251, cls=2.114
Time for epoch: 192.25 seconds
Updating best test loss: 2.51026

Starting epoch 43 / 50
Learning Rate for this epoch: 1e-05
Epoch [43/50], Iter [50/418], Loss: total=1.765, reg=10.314,
containing_obj=6.523, no_obj=2.315, cls=2.033
Epoch [43/50], Iter [100/418], Loss: total=1.852, reg=10.670,
containing_obj=6.945, no_obj=2.336, cls=2.269
Epoch [43/50], Iter [150/418], Loss: total=1.857, reg=10.676,
containing_obj=7.002, no_obj=2.309, cls=2.295
Epoch [43/50], Iter [200/418], Loss: total=1.860, reg=10.860,
containing_obj=6.989, no_obj=2.312, cls=2.164
Epoch [43/50], Iter [250/418], Loss: total=1.838, reg=10.716,
containing_obj=6.934, no_obj=2.304, cls=2.099
Epoch [43/50], Iter [300/418], Loss: total=1.838, reg=10.650,
containing_obj=6.968, no_obj=2.309, cls=2.131

Epoch [43/50], Iter [350/418], Loss: total=1.842, reg=10.707,
containing_obj=6.991, no_obj=2.290, cls=2.111
Epoch [43/50], Iter [400/418], Loss: total=1.845, reg=10.742,
containing_obj=7.006, no_obj=2.280, cls=2.107
Time for epoch: 192.59 seconds

Starting epoch 44 / 50

Learning Rate for this epoch: 1e-05

Epoch [44/50], Iter [50/418], Loss: total=1.809, reg=10.507,
containing_obj=6.727, no_obj=2.287, cls=2.187
Epoch [44/50], Iter [100/418], Loss: total=1.825, reg=10.659,
containing_obj=6.798, no_obj=2.268, cls=2.178
Epoch [44/50], Iter [150/418], Loss: total=1.842, reg=10.633,
containing_obj=6.966, no_obj=2.271, cls=2.235
Epoch [44/50], Iter [200/418], Loss: total=1.839, reg=10.558,
containing_obj=6.993, no_obj=2.294, cls=2.230
Epoch [44/50], Iter [250/418], Loss: total=1.856, reg=10.668,
containing_obj=7.071, no_obj=2.285, cls=2.250
Epoch [44/50], Iter [300/418], Loss: total=1.855, reg=10.679,
containing_obj=7.083, no_obj=2.286, cls=2.215
Epoch [44/50], Iter [350/418], Loss: total=1.853, reg=10.668,
containing_obj=7.095, no_obj=2.293, cls=2.184
Epoch [44/50], Iter [400/418], Loss: total=1.856, reg=10.708,
containing_obj=7.105, no_obj=2.285, cls=2.178
Time for epoch: 192.45 seconds

Starting epoch 45 / 50

Learning Rate for this epoch: 1e-05

Epoch [45/50], Iter [50/418], Loss: total=1.897, reg=10.579,
containing_obj=7.503, no_obj=2.307, cls=2.381
Epoch [45/50], Iter [100/418], Loss: total=1.834, reg=10.305,
containing_obj=7.214, no_obj=2.358, cls=2.127
Epoch [45/50], Iter [150/418], Loss: total=1.844, reg=10.390,
containing_obj=7.291, no_obj=2.332, cls=2.117
Epoch [45/50], Iter [200/418], Loss: total=1.857, reg=10.517,
containing_obj=7.285, no_obj=2.334, cls=2.146
Epoch [45/50], Iter [250/418], Loss: total=1.852, reg=10.550,
containing_obj=7.203, no_obj=2.346, cls=2.121
Epoch [45/50], Iter [300/418], Loss: total=1.850, reg=10.561,
containing_obj=7.149, no_obj=2.347, cls=2.145
Epoch [45/50], Iter [350/418], Loss: total=1.869, reg=10.757,
containing_obj=7.187, no_obj=2.343, cls=2.145
Epoch [45/50], Iter [400/418], Loss: total=1.869, reg=10.805,
containing_obj=7.150, no_obj=2.336, cls=2.140
Time for epoch: 192.30 seconds
---Evaluate model on test samples---

100%| | 4950/4950 [01:24<00:00, 58.55it/s]

```
---class aeroplane ap 0.5155657633773274---
---class bicycle ap 0.6407249854153532---
---class bird ap 0.5195222905042562---
---class boat ap 0.29956580337068556---
---class bottle ap 0.2016880944431818---
---class bus ap 0.6494355794363578---
---class car ap 0.6717643638533068---
---class cat ap 0.7149326443600601---
---class chair ap 0.28272703081501294---
---class cow ap 0.5803990867306565---
---class diningtable ap 0.36512023570516244---
---class dog ap 0.6581290541637009---
---class horse ap 0.6993358298319787---
---class motorbike ap 0.5894119383397063---
---class person ap 0.5065996918667861---
---class pottedplant ap 0.22934576300103537---
---class sheep ap 0.5124613674339418---
---class sofa ap 0.5381811754208683---
---class train ap 0.6696534829621963---
---class tvmonitor ap 0.5271823677311127---
---map 0.5185873274381344---
44 [np.float64(0.5155657633773274), np.float64(0.6407249854153532),
np.float64(0.5195222905042562), np.float64(0.29956580337068556),
np.float64(0.2016880944431818), np.float64(0.6494355794363578),
np.float64(0.6717643638533068), np.float64(0.7149326443600601),
np.float64(0.28272703081501294), np.float64(0.5803990867306565),
np.float64(0.36512023570516244), np.float64(0.6581290541637009),
np.float64(0.6993358298319787), np.float64(0.5894119383397063),
np.float64(0.5065996918667861), np.float64(0.22934576300103537),
np.float64(0.5124613674339418), np.float64(0.5381811754208683),
np.float64(0.6696534829621963), np.float64(0.5271823677311127)]
```

Starting epoch 46 / 50

Learning Rate for this epoch: 1e-05

```
Epoch [46/50], Iter [50/418], Loss: total=1.830, reg=10.526,
containing_obj=7.227, no_obj=2.314, cls=1.895
Epoch [46/50], Iter [100/418], Loss: total=1.823, reg=10.566,
containing_obj=7.098, no_obj=2.322, cls=1.894
Epoch [46/50], Iter [150/418], Loss: total=1.826, reg=10.641,
containing_obj=7.015, no_obj=2.341, cls=1.910
Epoch [46/50], Iter [200/418], Loss: total=1.845, reg=10.731,
containing_obj=7.018, no_obj=2.313, cls=2.081
Epoch [46/50], Iter [250/418], Loss: total=1.855, reg=10.772,
containing_obj=7.061, no_obj=2.322, cls=2.100
Epoch [46/50], Iter [300/418], Loss: total=1.864, reg=10.803,
```

containing_obj=7.085, no_obj=2.330, cls=2.152
Epoch [46/50], Iter [350/418], Loss: total=1.871, reg=10.891,
containing_obj=7.068, no_obj=2.323, cls=2.173
Epoch [46/50], Iter [400/418], Loss: total=1.861, reg=10.823,
containing_obj=7.057, no_obj=2.314, cls=2.137
Time for epoch: 192.41 seconds
Updating best test loss: 2.50839

Starting epoch 47 / 50

Learning Rate for this epoch: 1e-05
Epoch [47/50], Iter [50/418], Loss: total=2.030, reg=12.086,
containing_obj=7.742, no_obj=2.184, cls=2.347
Epoch [47/50], Iter [100/418], Loss: total=1.925, reg=11.370,
containing_obj=7.213, no_obj=2.255, cls=2.267
Epoch [47/50], Iter [150/418], Loss: total=1.896, reg=11.180,
containing_obj=7.155, no_obj=2.251, cls=2.167
Epoch [47/50], Iter [200/418], Loss: total=1.909, reg=11.331,
containing_obj=7.197, no_obj=2.274, cls=2.104
Epoch [47/50], Iter [250/418], Loss: total=1.870, reg=11.039,
containing_obj=7.040, no_obj=2.284, cls=2.073
Epoch [47/50], Iter [300/418], Loss: total=1.877, reg=11.080,
containing_obj=7.057, no_obj=2.293, cls=2.095
Epoch [47/50], Iter [350/418], Loss: total=1.870, reg=10.992,
containing_obj=7.054, no_obj=2.307, cls=2.082
Epoch [47/50], Iter [400/418], Loss: total=1.866, reg=10.975,
containing_obj=7.055, no_obj=2.297, cls=2.070
Time for epoch: 192.28 seconds

Starting epoch 48 / 50

Learning Rate for this epoch: 1e-05
Epoch [48/50], Iter [50/418], Loss: total=1.754, reg=10.109,
containing_obj=6.677, no_obj=2.301, cls=1.956
Epoch [48/50], Iter [100/418], Loss: total=1.791, reg=10.305,
containing_obj=6.768, no_obj=2.332, cls=2.091
Epoch [48/50], Iter [150/418], Loss: total=1.792, reg=10.399,
containing_obj=6.808, no_obj=2.319, cls=1.974
Epoch [48/50], Iter [200/418], Loss: total=1.796, reg=10.416,
containing_obj=6.833, no_obj=2.313, cls=1.987
Epoch [48/50], Iter [250/418], Loss: total=1.813, reg=10.547,
containing_obj=6.957, no_obj=2.290, cls=1.959
Epoch [48/50], Iter [300/418], Loss: total=1.827, reg=10.653,
containing_obj=6.997, no_obj=2.276, cls=1.993
Epoch [48/50], Iter [350/418], Loss: total=1.854, reg=10.864,
containing_obj=7.094, no_obj=2.274, cls=2.010
Epoch [48/50], Iter [400/418], Loss: total=1.851, reg=10.853,
containing_obj=7.050, no_obj=2.286, cls=2.019

Time for epoch: 192.39 seconds

Starting epoch 49 / 50

Learning Rate for this epoch: 1e-05

Epoch [49/50], Iter [50/418], Loss: total=1.835, reg=10.388,
containing_obj=7.479, no_obj=2.239, cls=1.910
Epoch [49/50], Iter [100/418], Loss: total=1.817, reg=10.341,
containing_obj=7.219, no_obj=2.260, cls=1.986
Epoch [49/50], Iter [150/418], Loss: total=1.853, reg=10.663,
containing_obj=7.293, no_obj=2.255, cls=2.029
Epoch [49/50], Iter [200/418], Loss: total=1.850, reg=10.691,
containing_obj=7.262, no_obj=2.280, cls=1.967
Epoch [49/50], Iter [250/418], Loss: total=1.849, reg=10.716,
containing_obj=7.194, no_obj=2.293, cls=1.984
Epoch [49/50], Iter [300/418], Loss: total=1.855, reg=10.722,
containing_obj=7.185, no_obj=2.291, cls=2.064
Epoch [49/50], Iter [350/418], Loss: total=1.849, reg=10.676,
containing_obj=7.167, no_obj=2.281, cls=2.059
Epoch [49/50], Iter [400/418], Loss: total=1.845, reg=10.651,
containing_obj=7.132, no_obj=2.280, cls=2.081
Time for epoch: 191.09 seconds

Starting epoch 50 / 50

Learning Rate for this epoch: 1e-05

Epoch [50/50], Iter [50/418], Loss: total=1.914, reg=11.097,
containing_obj=7.216, no_obj=2.291, cls=2.364
Epoch [50/50], Iter [100/418], Loss: total=1.846, reg=10.723,
containing_obj=7.012, no_obj=2.306, cls=2.117
Epoch [50/50], Iter [150/418], Loss: total=1.869, reg=10.919,
containing_obj=7.053, no_obj=2.323, cls=2.136
Epoch [50/50], Iter [200/418], Loss: total=1.862, reg=10.938,
containing_obj=7.020, no_obj=2.285, cls=2.099
Epoch [50/50], Iter [250/418], Loss: total=1.875, reg=10.943,
containing_obj=7.151, no_obj=2.267, cls=2.139
Epoch [50/50], Iter [300/418], Loss: total=1.872, reg=10.927,
containing_obj=7.138, no_obj=2.289, cls=2.113
Epoch [50/50], Iter [350/418], Loss: total=1.866, reg=10.826,
containing_obj=7.135, no_obj=2.304, cls=2.122
Epoch [50/50], Iter [400/418], Loss: total=1.857, reg=10.823,
containing_obj=7.036, no_obj=2.304, cls=2.118
Time for epoch: 190.57 seconds

---Evaluate model on test samples---

100%| | 4950/4950 [01:24<00:00, 58.88it/s]

---class aeroplane ap 0.5286195038711792---

---class bicycle ap 0.6386598786845847---

```

---class bird ap 0.5144893662598363---
---class boat ap 0.2864402653839466---
---class bottle ap 0.19790010399037983---
---class bus ap 0.6387810998162872---
---class car ap 0.6665124802647417---
---class cat ap 0.687172237588932---
---class chair ap 0.28968744850955425---
---class cow ap 0.5662775612781514---
---class diningtable ap 0.3555324946888925---
---class dog ap 0.6600585709436384---
---class horse ap 0.6975130206386128---
---class motorbike ap 0.5866140673331507---
---class person ap 0.497727692596479---
---class pottedplant ap 0.23192333275058358---
---class sheep ap 0.5119559272838976---
---class sofa ap 0.5380399233741392---
---class train ap 0.6675650036280616---
---class tvmonitor ap 0.5135638830925355---
---map 0.5137516930988792---
49 [np.float64(0.5286195038711792), np.float64(0.6386598786845847),
np.float64(0.5144893662598363), np.float64(0.2864402653839466),
np.float64(0.19790010399037983), np.float64(0.6387810998162872),
np.float64(0.6665124802647417), np.float64(0.687172237588932),
np.float64(0.28968744850955425), np.float64(0.5662775612781514),
np.float64(0.3555324946888925), np.float64(0.6600585709436384),
np.float64(0.6975130206386128), np.float64(0.5866140673331507),
np.float64(0.497727692596479), np.float64(0.23192333275058358),
np.float64(0.5119559272838976), np.float64(0.5380399233741392),
np.float64(0.6675650036280616), np.float64(0.5135638830925355)]

```

```

[ ]: if False:
    LATEST_NUMBER = 10

    # Load the network weights from epoch 5 checkpoint before training
    checkpoint_path = f'checkpoints/detector_epoch_{LATEST_NUMBER}.pth'
    if os.path.exists(checkpoint_path):
        print(f"Loading checkpoint from {checkpoint_path}")
        net.load_state_dict(torch.load(checkpoint_path, map_location=device))
    else:
        print(f"Checkpoint {checkpoint_path} not found. Training from scratch.")

    best_test_loss = np.inf
    learning_rate = learning_rate

    os.makedirs('./checkpoints', exist_ok=True)
    for epoch in range(LATEST_NUMBER, num_epochs):
        net.train()

```

```

# Update learning rate late in training
if epoch == 30 or epoch == 40:
    learning_rate /= 10.0

for param_group in optimizer.param_groups:
    param_group['lr'] = learning_rate

print('\n\nStarting epoch %d / %d' % (epoch + 1, num_epochs))
print('Learning Rate for this epoch: {}'.format(learning_rate))

total_loss = collections.defaultdict(int)

start_iteration_time = time.time()
for i, data in enumerate(train_loader):
    data = (item.to(device) for item in data)
    images, target_boxes, target_cls, has_object_map = data
    pred = net(images)
    loss_dict = criterion(pred, target_boxes, target_cls,
↪has_object_map)
    for key in loss_dict:
        total_loss[key] += loss_dict[key].item()

    optimizer.zero_grad()
    loss_dict['total_loss'].backward()
    optimizer.step()

    if (i+1) % 50 == 0:
        outstring = 'Epoch [%d/%d], Iter [%d/%d], Loss: ' % ((epoch+1,
↪num_epochs, i+1, len(train_loader)))
        outstring += ', '.join( "%s=%.3f" % (key[:5], val / (i+1)) for
↪key, val in total_loss.items() )
        print(outstring)
        print('Time for epoch: %.2f seconds' % (time.time() -
↪start_iteration_time))

# evaluate the network on the test data
if (epoch + 1) % 5 == 0:
    test_aps = evaluate(net, test_dataset_file=annotation_file_test,
↪img_root=file_root_test)
    print(epoch, test_aps)
    with torch.no_grad():
        test_loss = 0.0
        net.eval()
        for i, data in enumerate(test_loader):
            data = (item.to(device) for item in data)
            images, target_boxes, target_cls, has_object_map = data

```

```

        pred = net(images)
        loss_dict = criterion(pred, target_boxes, target_cls,
↪has_object_map)
        test_loss += loss_dict['total_loss'].item()
        test_loss /= len(test_loader)

    if best_test_loss > test_loss:
        best_test_loss = test_loss
        print('Updating best test loss: %.5f' % best_test_loss)
        torch.save(net.state_dict(), 'checkpoints/best_detector.pth')

    if (epoch+1) in [5, 10, 15, 20, 25, 30, 35, 40]:
        torch.save(net.state_dict(), 'checkpoints/detector_epoch_%d.pth' %
↪(epoch+1))

    torch.save(net.state_dict(), 'checkpoints/detector.pth')

```

1 View example predictions

```

[39]: net.eval()

# select random image from test set
image_name = random.choice(test_dataset.fnames)
image = cv2.imread(os.path.join(file_root_test, image_name))
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

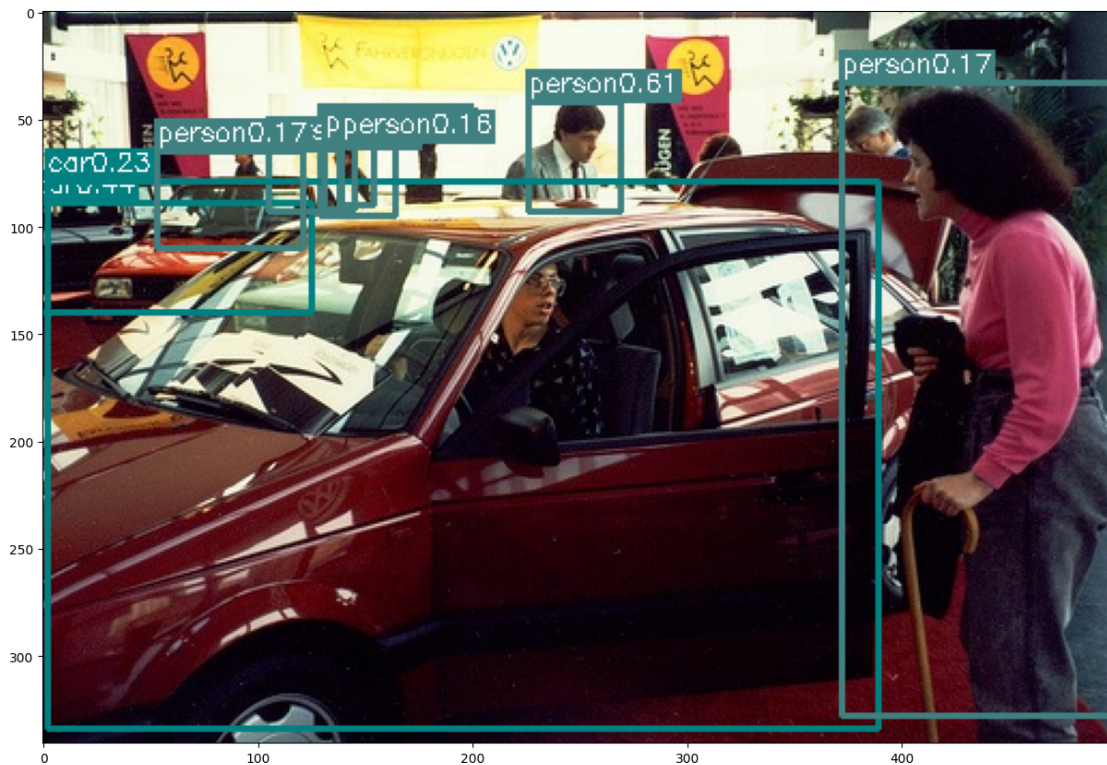
print('predicting...')
result = predict_image(net, image_name, root_img_directory=file_root_test)
for left_up, right_bottom, class_name, _, prob in result:
    color = COLORS[VOC_CLASSES.index(class_name)]
    cv2.rectangle(image, left_up, right_bottom, color, 2)
    label = class_name + str(round(prob, 2))
    text_size, baseline = cv2.getTextSize(label, cv2.FONT_HERSHEY_SIMPLEX, 0.4,
↪1)
    p1 = (left_up[0], left_up[1] - text_size[1])
    cv2.rectangle(image, (p1[0] - 2 // 2, p1[1] - 2 - baseline), (p1[0] +
↪text_size[0], p1[1] + text_size[1]),
        color, -1)
    cv2.putText(image, label, (p1[0], p1[1] + baseline), cv2.
↪FONT_HERSHEY_SIMPLEX, 0.4, (255, 255, 255), 1, 8)

plt.figure(figsize = (15,15))
plt.imshow(image)

```

predicting...

[39]: <matplotlib.image.AxesImage at 0x4f1f677d0>



1.1 Evaluate on Test

To evaluate detection results we use mAP (mean of average precision over each class)

```
[13]: # All epochs
# Load the best model checkpoint
best_model_path = 'checkpoints/detector.pth'
if os.path.exists(best_model_path):
    print(f"Loading best model from {best_model_path}")
    net.load_state_dict(torch.load(best_model_path, map_location=device))
    net.eval()
else:
    print(f"Best model checkpoint {best_model_path} not found.")

test_aps = evaluate(net, test_dataset_file=annotation_file_test,
img_root=file_root_test)
```

Loading best model from checkpoints/detector.pth

---Evaluate model on test samples---

100% | 4950/4950 [01:24<00:00, 58.39it/s]

```

---class aeroplane ap 0.5286195038711792---
---class bicycle ap 0.6386598786845847---
---class bird ap 0.5144893662598363---
---class boat ap 0.2864402653839466---
---class bottle ap 0.19790010399037983---
---class bus ap 0.6387810998162872---
---class car ap 0.6665124802647417---
---class cat ap 0.687172237588932---
---class chair ap 0.28968744850955425---
---class cow ap 0.5662775612781514---
---class diningtable ap 0.3555324946888925---
---class dog ap 0.6600585709436384---
---class horse ap 0.6975130206386128---
---class motorbike ap 0.5866140673331507---
---class person ap 0.497727692596479---
---class pottedplant ap 0.23192333275058358---
---class sheep ap 0.5119559272838976---
---class sofa ap 0.5380399233741392---
---class train ap 0.6675650036280616---
---class tvmonitor ap 0.5135638830925355---
---map 0.5137516930988792---

```

1.1.1 Cell added to get intermediate mAP values for students

```

[13]: network_paths = ['./checkpoints/detector_epoch_%d.pth' % epoch for epoch in [5,
    ↪10, 20, 30, 40]]+ ['./checkpoints/detector.pth']
for load_network_path in network_paths:
    print('Loading saved network from {}'.format(load_network_path))
    net_loaded = resnet50().to(device)
    net_loaded.load_state_dict(torch.load(load_network_path))
    evaluate(net_loaded, test_dataset_file=annotation_file_test,
    ↪img_root=file_root_test)

```

Loading saved network from ./checkpoints/detector_epoch_5.pth

---Evaluate model on test samples---

100%| | 4950/4950 [03:19<00:00, 24.86it/s]

```

---class aeroplane ap 0.0003189792663476874---
---class bicycle ap 0.0---
---class bird ap 0.0---
---class boat ap 0.0---
---class bottle ap 0.0---
---class bus ap 0.0---
---class car ap 0.0---
---class cat ap 1.6240093542938807e-05---
---class chair ap 0.0---
---class cow ap 0.0---
---class diningtable ap 0.0---

```

```

---class dog ap 0.0---
---class horse ap 0.0---
---class motorbike ap 0.0---
---class person ap 0.0030534143732986307---
---class pottedplant ap 0.0---
---class sheep ap 0.0---
---class sofa ap 5.0346058668866325e-06---
---class train ap 0.0---
---class tvmonitor ap 0.0---
---map 0.0001696834169528072---
Loading saved network from ./checkpoints/detector_epoch_10.pth
---Evaluate model on test samples---

100%|      | 4950/4950 [02:01<00:00, 40.84it/s]

---class aeroplane ap 0.3014771097780536---
---class bicycle ap 0.36129658040223733---
---class bird ap 0.2534141665673341---
---class boat ap 0.07343590168929827---
---class bottle ap 0.04227217257341699---
---class bus ap 0.17271955633441674---
---class car ap 0.4169200539376061---
---class cat ap 0.4512988517326965---
---class chair ap 0.15286722434864022---
---class cow ap 0.22241577841893784---
---class diningtable ap 0.028432732316227463---
---class dog ap 0.4464640386148201---
---class horse ap 0.48321168267725856---
---class motorbike ap 0.42621271511382786---
---class person ap 0.3307705571662599---
---class pottedplant ap 0.0510021749463492---
---class sheep ap 0.12924341380581822---
---class sofa ap 0.12966925785902456---
---class train ap 0.3250704121609788---
---class tvmonitor ap 0.2561439547886648---
---map 0.25271691676159336---
Loading saved network from ./checkpoints/detector_epoch_20.pth
---Evaluate model on test samples---

100%|      | 4950/4950 [01:59<00:00, 41.42it/s]

---class aeroplane ap 0.38373665045678396---
---class bicycle ap 0.5357289364138224---
---class bird ap 0.34199662216086213---
---class boat ap 0.19278342433626366---
---class bottle ap 0.13427908521244628---
---class bus ap 0.5095167743278147---
---class car ap 0.6028684117613615---
---class cat ap 0.5345679721047478---
---class chair ap 0.2414396155258963---

```

```

---class cow ap 0.3833701882162965---
---class diningtable ap 0.2905840139183931---
---class dog ap 0.6156815830818944---
---class horse ap 0.6450003304307534---
---class motorbike ap 0.47418563162654603---
---class person ap 0.451757499035337---
---class pottedplant ap 0.11735358711965518---
---class sheep ap 0.3620237782139003---
---class sofa ap 0.44845426260646637---
---class train ap 0.5145580299886048---
---class tvmonitor ap 0.3798173970925678---
---map 0.40798518968152064---
Loading saved network from ./checkpoints/detector_epoch_30.pth
---Evaluate model on test samples---

100%|      | 4950/4950 [01:59<00:00, 41.32it/s]

---class aeroplane ap 0.42918222030881---
---class bicycle ap 0.6205642510255927---
---class bird ap 0.4471715379161858---
---class boat ap 0.2587868237142986---
---class bottle ap 0.17507680789311492---
---class bus ap 0.5893322340909688---
---class car ap 0.6497255806086264---
---class cat ap 0.6738586006640961---
---class chair ap 0.25414065272068614---
---class cow ap 0.5156951850644542---
---class diningtable ap 0.34566773606517404---
---class dog ap 0.6403550895129018---
---class horse ap 0.6654910046936673---
---class motorbike ap 0.5576233402742758---
---class person ap 0.4872254592979751---
---class pottedplant ap 0.1679885050521627---
---class sheep ap 0.49208715898819116---
---class sofa ap 0.4785689345066969---
---class train ap 0.6550972961108075---
---class tvmonitor ap 0.47834335800505523---
---map 0.47909908882568714---
Loading saved network from ./checkpoints/detector_epoch_40.pth
---Evaluate model on test samples---

100%|      | 4950/4950 [01:48<00:00, 45.65it/s]

---class aeroplane ap 0.5454054394869845---
---class bicycle ap 0.6221017178885075---
---class bird ap 0.5169982257513247---
---class boat ap 0.30084837128500574---
---class bottle ap 0.20398418726967044---
---class bus ap 0.6347290978950576---
---class car ap 0.6779320696243094---

```

```

---class cat ap 0.698474195009899---
---class chair ap 0.2879846474477097---
---class cow ap 0.5422101372053073---
---class diningtable ap 0.37498605595537826---
---class dog ap 0.6536593565219815---
---class horse ap 0.6949596783568934---
---class motorbike ap 0.5878443833189635---
---class person ap 0.49499801795753773---
---class pottedplant ap 0.24380271222725589---
---class sheep ap 0.4964193389150033---
---class sofa ap 0.5270604121211506---
---class train ap 0.6609148871902215---
---class tvmonitor ap 0.5087950549610453---
---map 0.5137053993194604---
Loading saved network from ./checkpoints/detector.pth
---Evaluate model on test samples---

100%|      | 4950/4950 [02:46<00:00, 29.76it/s]

---class aeroplane ap 0.0---
---class bicycle ap 0.0---
---class bird ap 0.0---
---class boat ap 7.75081738684164e-07---
---class bottle ap 0.0---
---class bus ap 0.0---
---class car ap 2.3267721088833185e-05---
---class cat ap 0.0---
---class chair ap 2.7272615215348724e-07---
---class cow ap 0.0---
---class diningtable ap 6.292933539070306e-07---
---class dog ap 0.0---
---class horse ap 0.0---
---class motorbike ap 0.0---
---class person ap 0.00856813171685639---
---class pottedplant ap 0.0---
---class sheep ap 0.0---
---class sofa ap 0.0---
---class train ap 0.0---
---class tvmonitor ap 3.436445011381506e-07---
---map 0.0004296710091845553---

```

```
[14]: output_submission_csv('my_solution.csv', test_aps)
```

2 Extra Credit

```
[ ]: from tqdm import tqdm

# select random image from test set
image_folder = 'extra-credit/video/'
net.eval()

# Ensure output directory exists
output_dir = os.path.join(image_folder, 'output')
os.makedirs(output_dir, exist_ok=True)

# Process images
for id in tqdm(range(1, 2148), desc="Processing images"):
    image_name = f'out{id}.png'
    image_name_out = f'out{id:04d}.png'
    image_path = os.path.join(image_folder, image_name)
    output_path = os.path.join(output_dir, image_name_out)
    image = cv2.imread(image_path)
    if image is None:
        print(f"Warning: Unable to read {image_name}. Skipping.")
        continue
    image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

    # Predict and draw bounding boxes
    result = predict_image(net, image_name, root_img_directory=image_folder)
    for left_up, right_bottom, class_name, _, prob in result:
        color = COLORS[VOC_CLASSES.index(class_name)]
        cv2.rectangle(image, left_up, right_bottom, color, 2)
        label = f"{class_name}{round(prob, 2)}"
        text_size, baseline = cv2.getTextSize(label, cv2.FONT_HERSHEY_SIMPLEX,
↪0.4, 1)
        p1 = (left_up[0], left_up[1] - text_size[1])
        cv2.rectangle(image, (p1[0] - 2 // 2, p1[1] - 2 - baseline),
                        (p1[0] + text_size[0], p1[1] + text_size[1]), color, -1)
        cv2.putText(image, label, (p1[0], p1[1] + baseline),
                    cv2.FONT_HERSHEY_SIMPLEX, 0.4, (255, 255, 255), 1, 8)

    # Save the image with bounding boxes
    image = cv2.cvtColor(image, cv2.COLOR_RGB2BGR)
    cv2.imwrite(output_path, image)

print("Processing complete!")
```